

Group 30: Final Report
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Bound-Less Crutches

A Portable, Dynamically Loaded Crutch Design

MECHENG 3E05
Project Mentor: Dr. Fatemeh Jazinizadeh

Team Member	Student ID
Elise Schwarz	
Taylor Kramer	
Meera Moorthy	
Nathan Urbshott	
Aiden Delaney	

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Project Management

During this project, the team met twice weekly with additional meetings decided on as needed. One of the weekly meetings was intended to be an intra-group meeting where the team could decide on action items for the week, complete work objectives, and prepare presentations for the second weekly meeting. The second weekly meetings included the faculty mentor and were intended to keep the project on course and ensure that the team deliverables were in-line with the course objectives. Meeting minute documentation and meeting to do list construction responsibilities were divided between team members in a weekly cycle, with every team member sharing in the managerial tasks.

Team members were not given explicit roles, but each adapted to a particular specialty in developing one or two components of the project in greater depth. CAD design and printing, mathematical model development, mathematical stress analysis, and financial analysis were each addressed by one or more team members with an elevated level of dedication. Progress was brought to weekly meetings where the next steps and adjustments could be determined by the group so that no one section of the project would diverge from the others.

A Gantt chart was created and updated throughout the project, which can be seen as an appended file. This chart was intended to keep the group aware of deadlines, which were adjusted at team meetings based on group consensus. Additional milestones and deadlines were created apart from those given in the course outline to allow for proper pacing of the project and not lead to a time crunch at any point in the development process. Alternate plans were also laid out to avoid that eventuality, and the team frequently reassessed its position in order to make sure that adjustments were being made to ensure the delivery of objectives. Some of the design objectives would have been eliminated to ensure the completion of the remaining project deliverables, but this was not necessary, and all project goals were accomplished.

Budgeting was integrated at every level of the project, where material selection and part selection were assessed as development proceeded. This budgetary analysis can be seen in the *Cost Analysis* section of this report.

Background Research

Axillary crutches are commonly used by people of all ages to aid in ambulation. The purpose of crutches is to relieve the weight-bearing load on an injured lower limb and transfer it to the torso and the arms by leaning on a crutch under each arm [1]. Axillary crutches can be needed for a variety of conditions, including sprains/strains/fractures in the leg, recovering after operative procedures, and other disabilities [1]. A standard axillary crutch is shown in Figure 1.

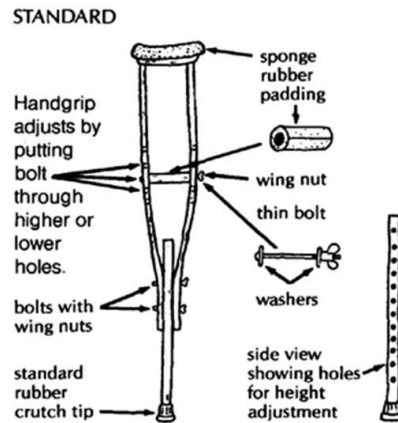


Figure 1: Standard axillary crutch with labels [2]

Traditional axillary crutches are entirely rigid structures; ground reaction forces at the feet of the crutches are transmitted directly to the components of the body they interact with. Previous studies have found that the greatest joint reaction forces can be observed in the wrist and shoulder, where the body is supported by the crutch [3]. These forces result in negative physiological impacts on underlying structures that are not adequately protected, such as nerves. “Crutch palsy” is a common peripheral neuropathy that develops as a result of prolonged compression of the **radial nerve** at the shoulder [4]. The **median and ulnar nerves** of the hand are also at risk of prolonged compression [5]. Nerve palsy can result in numbness, reduced dexterity and limb strength, or a complete inability to control muscles innervated by the affected nerve [6]. Treatment for crutch-related palsy is to stop the usage of crutches or remove the source of compression [7]. Although most patients recover completely, the treatment plan neglects to consider those who develop negative physiological effects from crutches but **do not have a mobility alternative**. An effective health solution should therefore consider mitigating the risk of **developing** the condition.

Radial, ulnar, and median nerve palsies are exacerbated by the **duration** of the compression as well as the **magnitude of force**. Previous studies have demonstrated that the addition of spring or spring-damper systems have the potential to reduce the overall reaction force, as well as the impulse (rate of change of force) [8].

Although there have been many advancements in crutches since they were invented over a century ago, there are still missing needs that are unfilled by crutches [9]. Specifically, due to the size, structure, and nature of axillary crutches, they are not easily collapsible, portable, or adaptable to different environments. This is a major inconvenience for regular crutch users, as they cannot be easily stowed away while in the car or while sitting down. It would be extremely advantageous if a pair of crutches could be packed away in a backpack for travel purposes, like to bring on a hike in case of an injury. Furthermore, the base of an axillary crutch is typically a flat rubber tip, which is often not compatible to terrain that is not a flat indoor surface. Since this rubber tip does not have a lot of traction, using crutches in conditions such as ice, sand, or gravel can pose a safety risk to the user.

Aside from the standard axillary crutches that are typically known and used in society, there have been recent crutch designs and experiments that have incorporated novel components. One current portable crutch in the market is the *Carex Folding Crutches*, which retail for \$47.95 [10]. The mechanism used to assemble and disassemble the crutch is shown below, in Figure 2. The primary limitation of this mechanism

is that it requires parts to completely taken apart and re-assembled, limiting the **speed** with which a user can convert the crutch between a portable and functional form.

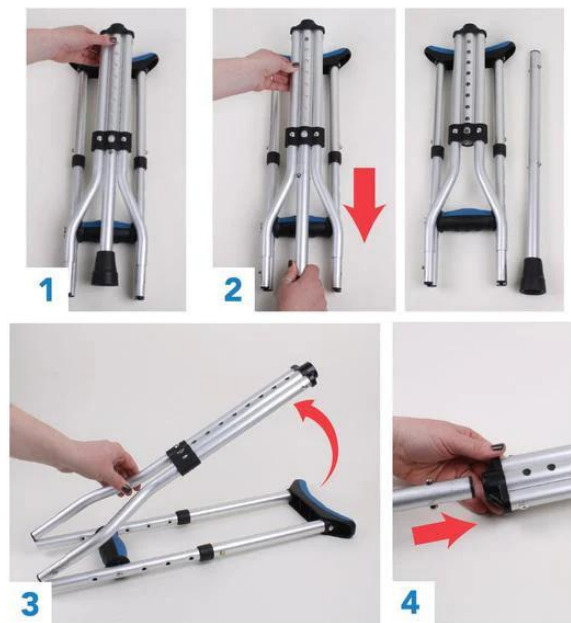


Figure 2: Portability mechanism for Carex Folding Crutches [10]

Another portable crutch available is the *Drive Medical Easy Adjust Universal Crutches* [11]. This design can be dissembled into two separate pieces for easy transport. The primary limitation of this design is its **portability**; in the portable state, the pieces are not constrained to each other. This raises the risk of transporting the crutch and potentially losing a part.



Figure 3: Portability mechanism for Drive Medical Easy Adjust Universal Crutches [11]

There are existing crutches which implement springs to reduce impulses applied to the body such as the *Adjustable Spring Cushion Crutches* made by York Nordic [12]. Since none of these products are

damped, they all have the same limitations of applying consistently high pressures to the underarm. They can also be unsafe when the spring unloads and forces the crutch's foot to slide prematurely.



Figure 4: Spring implementation in York Nordic's Adjustable Spring Cushion Crutches [12]

Problem Definition

With all the background information laid out, the requirement for changes to modern axillary crutch design becomes apparent. There is a **defined need** in the market for solutions to crutch related maladies and inconveniences. Specifically, a solution is required to eliminate crutch palsy by reducing the force and impact applied to the radial nerve, while also improving the crutch's portability and maintaining its ease of use and assembly so that it can be easily deployed in any situation.

Objective Statement

Our objectives can be split into two primary categories: design objectives and analysis objectives.

Design objectives will be used to evaluate the ability of the device to meet its desired function. The design objectives are as follows:

- Design a structure which will accommodate a selected range of body shapes and sizes
- Identify/design a mechanism to fold a crutch into a portable and compact form. The mechanism must be:
 - Intuitive and easy for users with limited mobility
 - Fast to assemble and disassemble (under 1 minute)
- Select/design a spring-damper system which:
 - Reduces the maximum applied force applied under the arm and at the hand during a user's full stride
 - Reduces the rate of increase and decrease of the applied force (impulse) through a stride
- Select appropriate materials throughout the product which are safe, lightweight, and durable

Analysis objectives will be used to evaluate the appropriate application of course concepts. The analysis objectives for this project are as follows:

- Measure the body momentum before/during/after contact (dynamic loading)
- Measure the impulse felt at the hand and shoulder (dynamic loading)
- Identify regions of stress concentration and critical points in the design
- Determine the effects of cyclical loading and fatigue on materials selected
- Determine the failure modes of the design in during both intended and unintended use cases
- Determine the effects of cyclical loading and fatigue on materials selected
- Determine failure criteria and compare with our chosen safety limits

Factors such as cost and manufacturing were also considered later, but these were the primary focus in the early development stage that led to the core tenets of the design.

Design Concept Selection Criteria

Table 1 below identifies and explains each of our selected design criteria. Our criteria are split into 3 distinct categories, as follows:

1. **User-based criteria:** these criteria address objectives related to the **use** of the device and its ability to perform the desired function.
2. **Design principle-based criteria:** these criteria address objectives related to the **application of design** principles
3. **Marketability-based criteria:** these criteria address the ability to market and ultimately sell the device.

Table 1: Design criteria explanation

Criteria	Explanation
User-Based Criteria	
Ease of use	Refers to the mechanism used to make the device portable. The design must be 'easy' and intuitive to assemble and disassemble. The easier to use, the higher the overall score.
Speed of assembly	Refers to the time required to transform the device between functional and portable forms. The faster it can be assembled, the higher the overall score.
Weight	Refers to the overall weight of the device and is an aspect of portability The heavier the device, the lower the overall score.
Design Principle-Based Criteria	
Ease of manufacturing	Refers to the manufacturing requirements for the device such as the use of standardized parts (higher rating) or the need for machining with high tolerances (lower rating)
Complexity of design	Refers to the variety of different stresses in the design and complexity of failure analysis. Designs with rotating parts, curved beams, or non-uniform cross-sections are rated higher.
Ease of assembly	Refers to the degree to which assembly can be automated and not require the hand-assembly.
Marketability-Based Criteria	
Cost	Refers to the cost of the device with reference to the complexity of the design, types of parts, manufacturing requirements, and assembly requirements.

Ease of repair	Refers to the ability to replace or repair parts that may fail without compromising the entire device.
Versatility	Refers to the diversity of users who can use the device (those with dexterity issues) and the overall portability of the device.
Aesthetics	Refers to the visual aesthetics of the device.

In order to weight the score for each criteria in a decision matrix, the criteria were ranked from most to least important. The most important criteria were **user-based** since they are a direct indicator of how well the device functions and its overall ability to serve its purpose. Within this, ease of use was ranked first, followed by speed of assembly, and then weight.

The next most important criteria were the **design principle-based** criteria since these consider the feasibility of the design with course constraints. Within this, manufacturing was ranked first, followed by complexity, and then assembly.

The least important criteria were the **marketability-based** criteria since these do not affect the function or design of the device, but impact the demographics that may purchase it. Within this, Cost was ranked first, followed by ease of repair, versatility, and then aesthetics.

The overall ranking of criteria is as follows:

1. Ease of use
2. Speed of assembly
3. Weight
4. Manufacturing
5. Complexity
6. Assembly
7. Cost
8. Ease of repair
9. Versatility
10. Aesthetics

Design Alternatives

When beginning the design process for our collapsible, dynamically loaded crutch, it was determined that there were many degrees of variability for how each mechanism required to achieve our design objectives could be created. We determined that the greatest degree of variation was present for the collapsible mechanism, and began developing various designs to address this problem.

DESIGN ALTERNATIVE #1: COLLAPSIBLE TRUSS

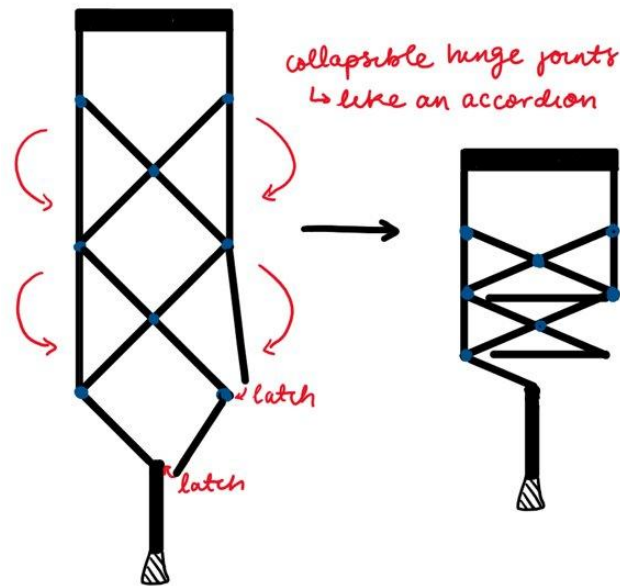


Figure 5: Collapsible truss folding mechanism for easily portable crutches

A collapsible truss structure was the first proposed folding mechanism for the crutch design and is shown above in Figure 5. This mechanism included hinged joints that would allow for the crutch to collapse and compress, similar to the mechanism by which an accordion works. This design would allow for the crutch to become very compact, making it much more portable than a standard crutch. Additionally, it was determined that this design would require two poles that can serve as latches to allow for the folding to occur, and for the crutch to lock in place when in the fully extended position.

Some advantages of this design would be that the crutch completely compresses into one unit, which would allow it to be very easy to pack into a vehicle or bag when traveling. Furthermore, since trusses are known to be extremely strong, this design would likely be very safe and capable of bearing additional weight. Some disadvantages of this mechanism would be the high number of failure points due to the many hinged joints, and the feasibility. Design iteration and research would need to be done to determine that the structure is sufficiently stable through the latching mechanism, and that it would not collapse upon impact. Additionally, placement for a handlebar would need to be decided, as the truss internals could create some degree of interference.

Relevant mechanical engineering concepts to this design include trusses, two member joints, hinged joints, combined loading, dynamic loading, stress concentration at joints, design failure.

DESIGN ALTERNATIVE #2: TENT POLES

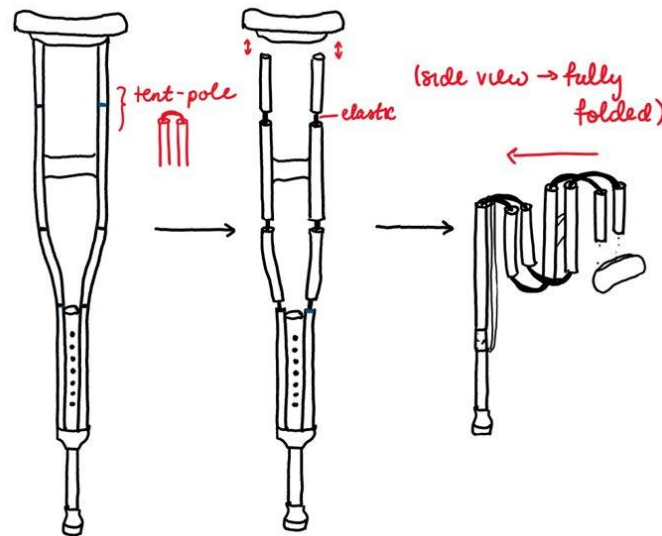


Figure 6: Collapsible tent pole folding mechanism for easily portable crutches

A collapsible tent pole structure was the second proposed folding mechanism for the crutch design, shown in Figure 6 above. This design includes a crutch that has three separation points, which are connected via internal elastics, similar to the structure of a tent pole. The elastic would be continuous throughout the length of the crutch, connecting all assembling pieces. This means that the internal of each crutch pole would be hollow, and the crutch would come apart by detaching the components and folding each section together through the elastic. In order for the pieces to come together, a locking mechanism would be required to ensure that the pieces cannot easily come apart when bearing load.

Some advantages of this design would be that it is very simple for the user to understand. Unlike the truss mechanism, this design does not largely change the overall appearance of the crutch, and so user interaction likely would not change much. Additionally, it is a familiar structure to a tent pole, so it would likely be easy for the user to understand. Some disadvantages of this structure are that a locking mechanism would need to be designed to keep the pieces in place. This mechanism could be challenging to implicate. Additionally, although collapsible, the design would be in pieces after taking apart, which is less ideal than the simple unit that the truss mechanism provided. A bag or other holding device would likely be required to house the components when taken apart.

Relevant mechanical engineering concepts in this design: rotating joints, stress concentration at joints, combined loading, dynamic loading, design for fatigue, design for failure.

DESIGN ALTERNATIVE #3: TELESCOPING POLES

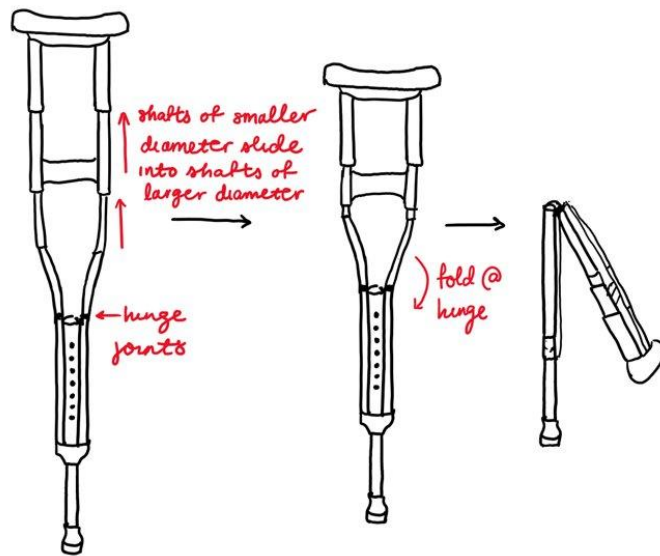


Figure 7: Collapsible telescoping pole folding mechanism for easily portable crutches

A collapsible telescoping pole structure was the third and final proposed folding mechanism for the crutch design, shown in Figure 7 above. This design includes telescoping poles that collapse internally to allow the structure to shrink down to a smaller size. It also includes a hinge mechanism to allow the structure to fold axially after telescoping inwards, allowing it to be more compact and easily portable. The telescoping poles would be actuated through a spring-loaded pin mechanism, allowing the user to extend and collapse the assembly through just the push of a button very easily.

Some advantages of this design are that it is fairly user intuitive. Many telescoping structures exist, and spring-loaded pins are a familiar design that many people interact with in their day-to-day lives. Additionally, this would allow the crutch to become extremely compact and easy to store, as there are no loose components since everything is stored inwards and then folded. It would also inadvertently allow for the crutch to become increasingly height adjustable, due to the telescoping nature. Some disadvantages of this design are that the telescoping poles could get bulky and add additional weight, making the device heavier. Heavy devices are not ideal for assistive technology, so this could be a major challenge. Also, the spring-loaded pins could potentially be load-bearing in a telescoping structure, which is not an ideal for area of stress localization.

Relevant mechanical engineering concepts in this design: sliding joints, hinged joints, stress concentration at joints, combined loading, dynamic loading, design for fatigue, design for failure.

After discussing each design along with its strengths and weaknesses, a decision matrix was created to evaluate which design best achieved our objectives and constraints. This matrix was created by assigning ranks to the previously outlined critical design criteria, as discussed in *Design Concept Selection Criteria*.

Decision Matrix

Table 2: Decision Matrix Highlighting Each Potential Conceptual Design

		Design #1: Truss		Design #2: Tent Poles		Design #3: Telescope	
Criteria	Weight	Score	Weighted	Score	Weighted	Score	Weighted
Ease of use	10	3	30	7	70	8	80
Speed	9	4	36	7	63	7	63
Manufacture	7	7	49	3	21	2	14
Assembly	5	6	30	7	35	3	15
Ease of repair	3	8	24	3	9	6	18
Cost	4	5	20	3	12	3	12
Weight	8	2	16	7	56	3	24
Aesthetics	1	2	2	8	8	7	7
Complexity	6	3	18	7	42	7	42
Versatility	2	6	12	6	12	7	14
Total			237		328		289

Upon evaluation of the decision matrix, it was determined that the tent-pole design best achieved our design constraints. With a score of 328, it was selected as our final concept solution. This structure would later be iterated in the coming weeks of our design project.

Design Iteration

After deciding that the tent-pole mechanism would be used as the folding mechanism for our design, we outlined the criteria that we wanted our complete design to encompass and began performing iterations. There were a total of 4 additional concepts that we wanted our design to include. These components are listed below:

1. Spring-Damper Assembly
2. Locking Mechanism
3. Adjustable Bases for Various Terrain/Environmental Applications
4. Adjustable Preload Mechanism

We performed design iteration on the spring-damper assembly and the locking mechanism only. The adjustable bases and adjustable preload mechanism were simple design choices and did not require iteration. They will be further elaborated in the Design Choice section of this report.

SPRING-DAMPER ASSEMBLY

In our early phases of design, we had only included a spring in the base of the crutch. The intention of the spring was to prolong the time of impact so that the force between the crutch base and the ground occurred over a longer time period, and the force applied back to the user was less drastic. Upon participating in design review sessions with our mentor, we determined that adding a damper to the assembly would allow for additional force dissipation, and that way the impulse could be decreased, and the user would experience a smaller force upon impact, which would help to mitigate the symptoms of crutch palsy. We performed a literature search to confirm these findings [13].

Parallel Mechanism #1

Initially, we designed the spring and damper mechanism to be in parallel. We determined that a parallel arrangement would be superior to a series arrangement because the spring could then perform elastic reload of the damper after each successive compression. We first considered the arrangement shown below:

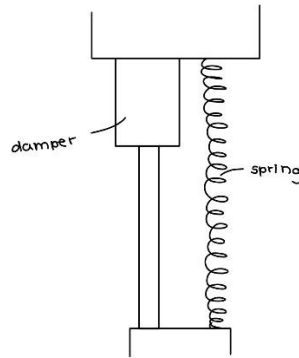


Figure 8: Initial Spring-Damper Mechanism

We ultimately determined that this arrangement was unstable and potentially could not bear large loads.

Parallel Mechanism #2

After performing some background research, we determined that it would be best to mimic the assembly of the spring-damper present in a car suspension. This assembly includes the damper internal to the spring, and the spring performs elastic reload of the damper after each successive compression and will be further explained in the Design Choice.

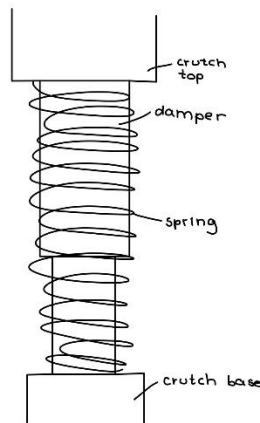


Figure 9: Spring-Damper Mechanism

LOCKING MECHANISM

Our team performed many iterations of the locking mechanism that would be used to keep the poles of the crutch attached. There were many design considerations that went into the final design of the locking mechanism, and our team experienced many challenges with selecting a mechanism that could achieve all the desired criteria.

The desired criteria for our locking mechanism was focused on three critical aspects:

1. Load bearing
 - Mechanism should not introduce any new points of stress concentration / potential failure
2. Single-handed, simultaneous actuation
 - The user should be able to actuate the mechanism using only one hand
 - The user must be able to detach both sides of the crutch simultaneously due to the mutual connection point of the poles at the armpit rest

Mechanism #1 – Twisting Mechanism

The first locking mechanism that our team designed was a twisting mechanism. This can be seen in Figure 10 below. This mechanism would work by the exterior pins on the bottom of the mechanism twisting into the lid of the top part of the mechanism, where they fit into a groove, and the elastic tying the pieces together would move through the central clearance of the mechanism.

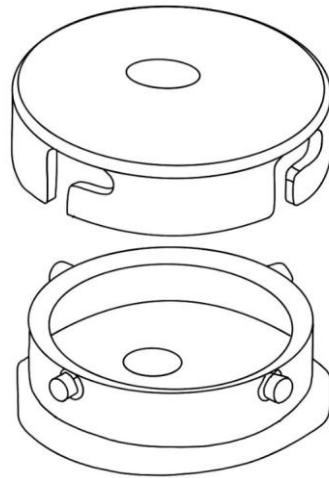


Figure 10: Initial Twisting Lock Mechanism

This mechanism was quickly rejected, as the twisting nature would be challenging to implement onto both sides of the crutch. Our team determined that it would be challenging to twist and untwist a mechanism for a number of reasons. Firstly, this would be challenging for the user to perform using only one hand. We wanted the user to be able to use only one hand so that they can take both sides of the crutch apart simultaneously, since it is a mirrored structure. Secondly, we determined that any twisting mechanism would be unfit. Since both crutch poles are attached to the foam armpit rest at the top of the crutch, a twisting mechanism on both sides would not be possible. After coming to this conclusion, our team focused on designing a locking mechanism that could be actuated linearly.

Mechanism #2 – Linear Ratchet Mechanism

The next proposed mechanism was a linear ratchet. This mechanism appeared superior to the previous, as it did not require any twisting/untwisting to occur: This mechanism is shown in Figure 11 and 12 below:

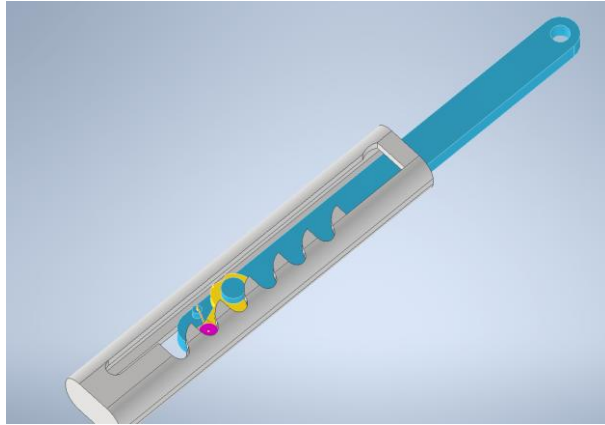


Figure 11: Linear Ratchet Mechanism

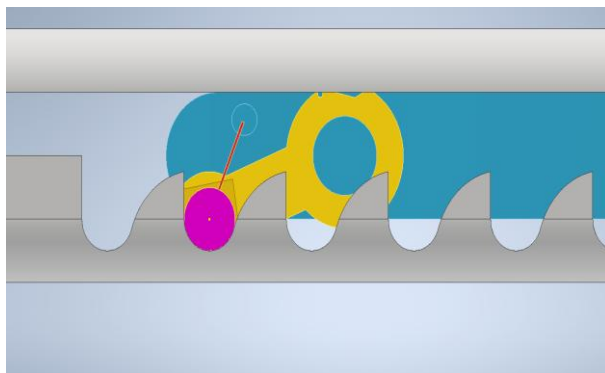


Figure 12: Linear Ratchet Mechanism

The idea behind this mechanism was that there would exist one ratchet at each interface of adjoining pieces, and the assembly could easily click together and click apart, keeping it shut tightly when in use but also easy to take apart. Ultimately, this design was rejected because it was determined that the mechanism would be load bearing, which would introduce a new position of stress concentration where potential failure could occur. We did not want to introduce any additional areas of stress concentration, so we decided to pursue a mechanism that would not be load bearing to provide less risk to the user.

Mechanism #3 – Spring Loaded Locking Mechanism

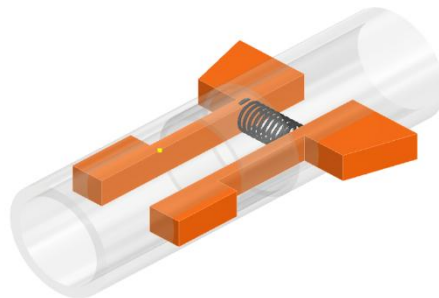


Figure 13: Linear Spring Loading Mechanism

The next designed mechanism included a spring-loaded mechanism, and also behaved linearly. The idea behind this mechanism would be that the user actuates the mechanism by pushing the orange components

inwards, which will work to compress the spring, and then as the spring compresses, a clearance is introduced at the bottom of the mechanism, where the orange component can completely slide out and allow the two pieces to detach. The main concern with this mechanism was that there was not a clearly provided path for the elastic to sit. We did not want the internal elastic to interfere with the spring compression, so we didn't see how this mechanism could be applied to our current tent pole design, which lead us to our fourth and final iteration.

Mechanism #4 – Dual Spring – Spring Loaded Locking Mechanism

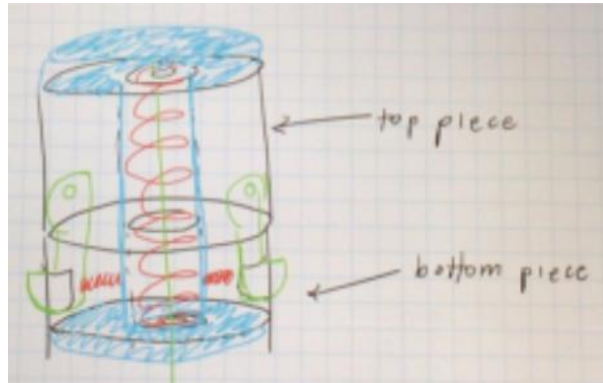


Figure 14: Dual Spring - Spring-Loaded Mechanism

The final locking mechanism that our team designed included two sets of springs. External springs connect the top and the bottom pieces together, through a spring-loaded push pin mechanism, so that the user can use one hand to separate the pieces. There is an internal spring present between the two pieces, which is compressed as the pieces are joined together. When the push pins are actuated, the internal spring is allowed to expand, causing the components to separate. This internal spring mechanism allowed room for the elastic to join the pieces. This was the final mechanism that our team chose to pursue, and few iterations were made from this point. This will be later elaborated in the Chosen Design component of this report.

Chosen Design

The design that we chose to pursue contains many different aforementioned elements that accomplish our design objectives. These components can be broken down into three sections:

1. Locking mechanism with elastics
2. Spring-damper system
3. Adjustable preload mechanism with adjustable bases

These components can be seen in the CAD model highlighted in Figure 15 below.

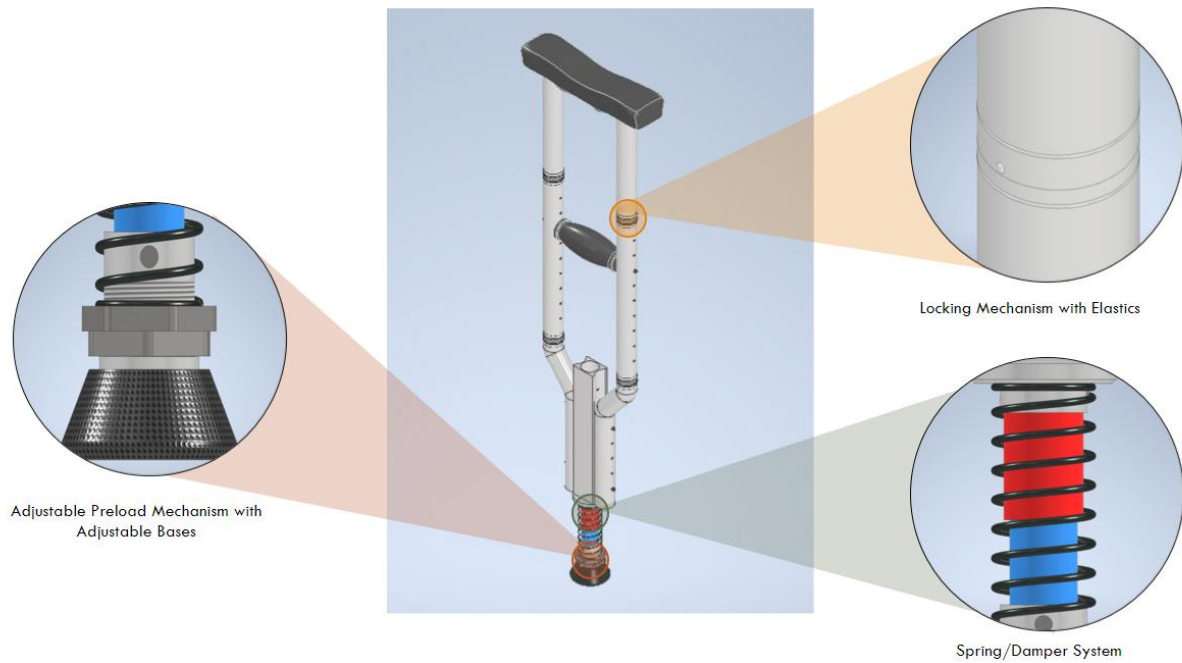


Figure 15: Complete Design

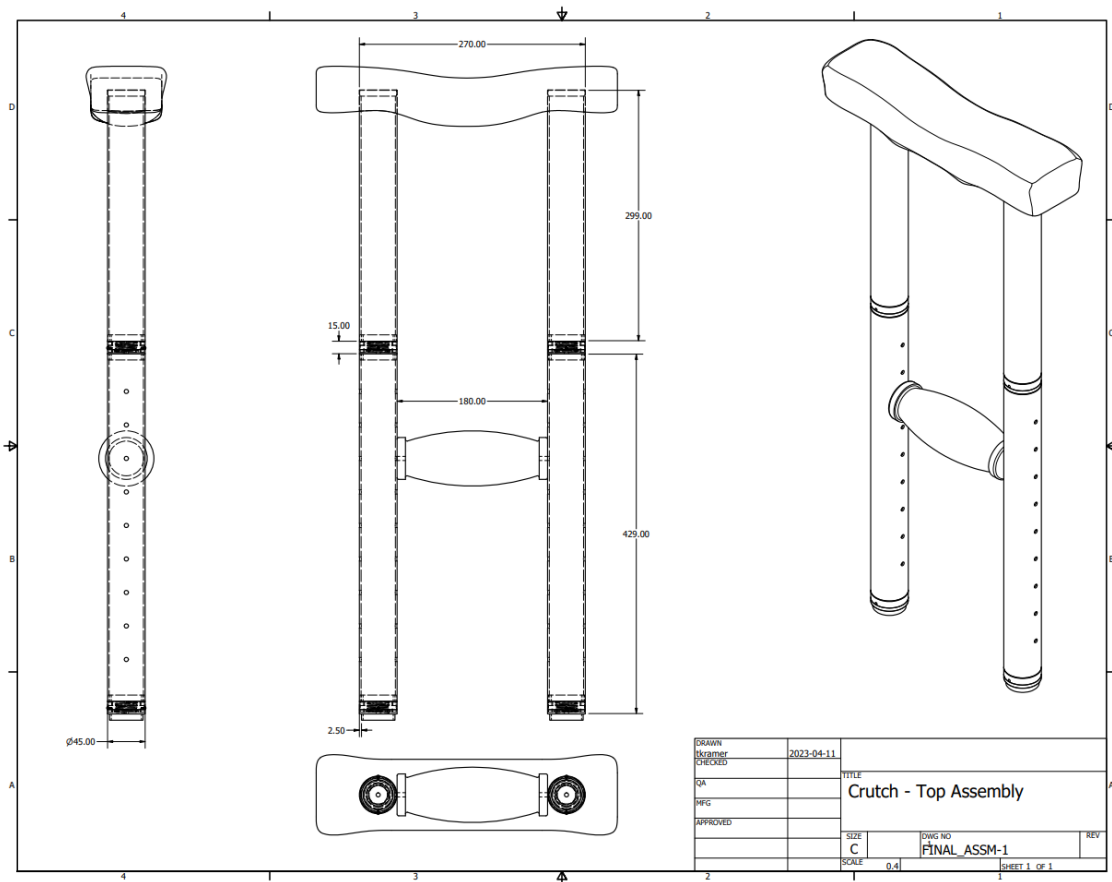


Figure 16: Engineering Drawing Highlighting Critical Dimensions of Crutch Top

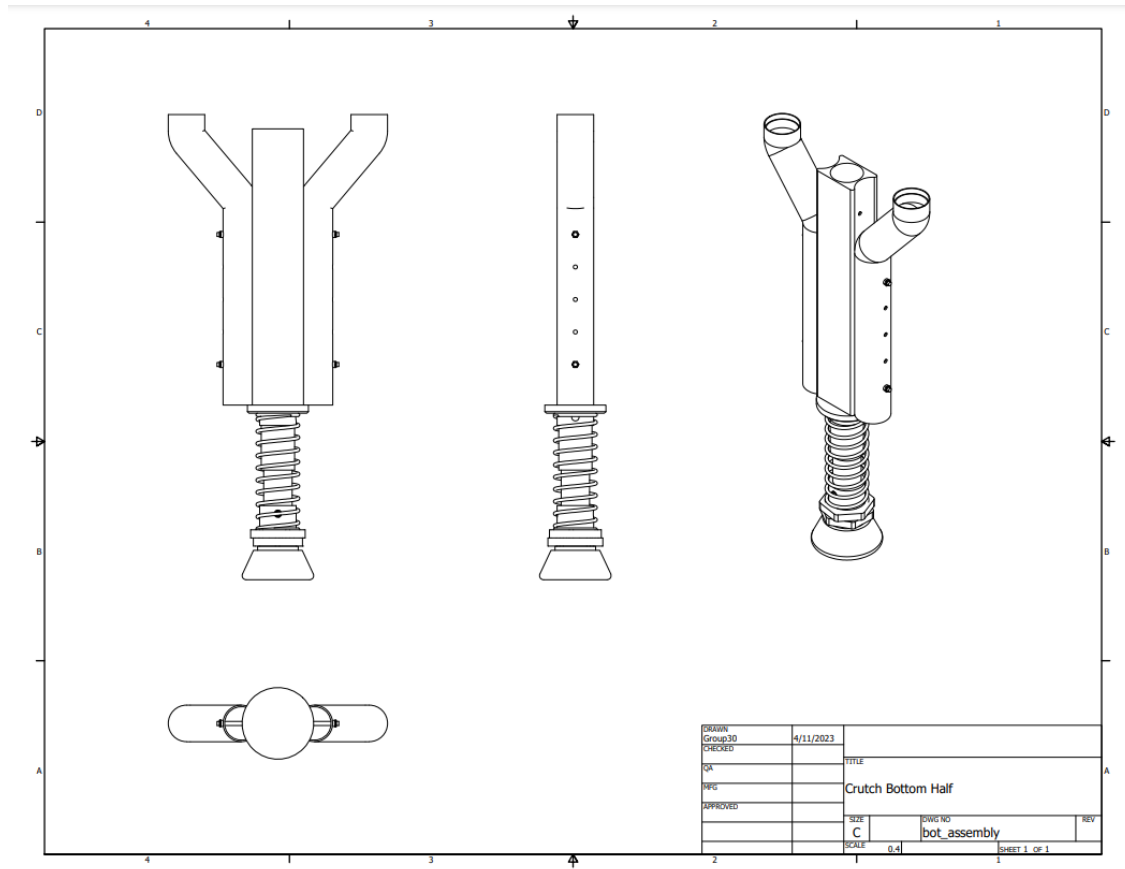


Figure 17: Engineering drawing highlighting the bottom half of the crutch

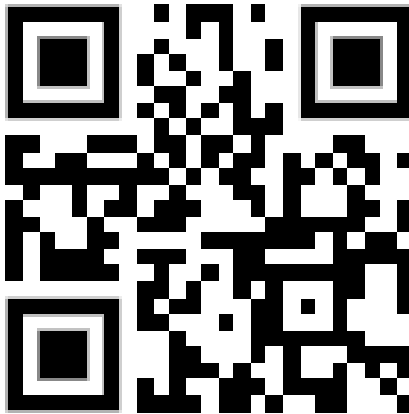


Figure 18: QR code of assembly simulation video

LOCKING MECHANISM WITH ELASTICS

The following locking mechanism was designed to be actuated by the user with one hand, so that the user can unclip both sides of the crutch at once. It was also designed to not be load bearing so that no additional points of stress concentration and potential failure were created. The mechanism is shown in Figure 19 below.

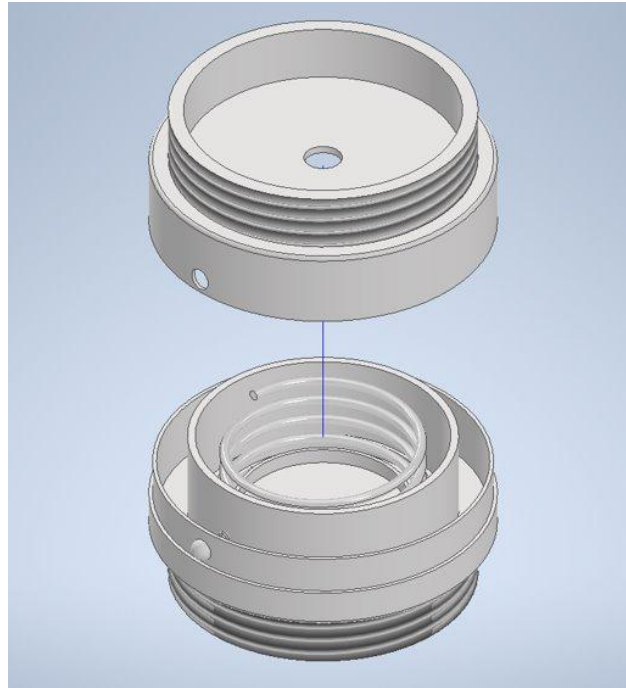


Figure 19: Locking Mechanism

This mechanism contains spring loaded pins in the bottom component which connect the top and bottom parts of the crutch together. There is a large internal spring housed within the mechanism that is compressed when the two components are locked shut. When the spring-loaded pins are engaged, the center spring is allowed to expand, which causes the top of the mechanism to shoot off the bottom. This mechanism also contains a clearance hole throughout to allow for space for the internal elastic. The spring-loaded pins push against a central cylinder to be engaged. The individual components are shown in the figure below.



Figure 20: Individual Components of Locking Mechanism

As seen above, these components are threaded into each part of the crutch pole to allow for easy attachment. It was critical for this design to be attached to the crutch poles instead of directly manufactured onto each pole in order to prevent manufacturing costs from becoming excessively large.

FOLDING MECHANISM

The locking mechanism is a critical component of the design as it allows for the crutch to be collapsible. Our design includes a total of 8 locking mechanisms, with 4 on each crutch and 2 on each side. The figure below highlights the folding of the crutch.

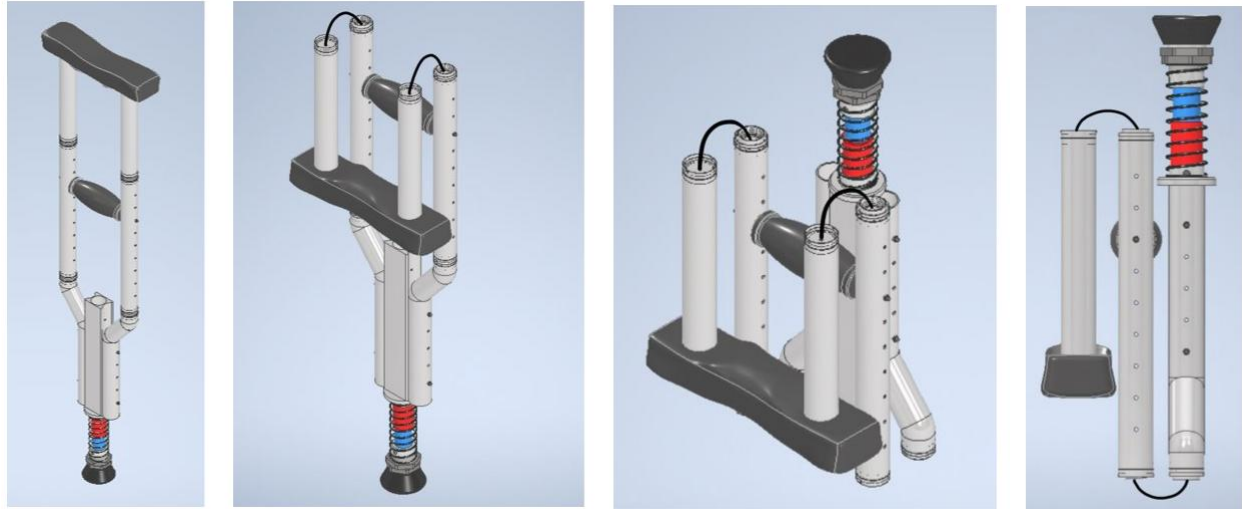


Figure 21: Folding Action of the Crutch

As seen from the image above, the bilateral locking mechanisms allow for the crutch to fold into three roughly equal components, making it easier to package and travel with.

SPRING-DAMPER SYSTEM

We included a spring-damper system in our assembly to allow for force dissipation in order to mitigate the symptoms of crutch palsy. The damper we selected, the MA225 miniature shock absorber from ACE, contains an internal viscoelastic oil that absorbs/dissipates energy in order to reduce the force applied on the user from the impact of the crutch base with the ground. The spring surrounds the damper to allow for elastic reload of the system. Upon each impact, the damper will compress and absorb energy, and when the applied force is released, the spring will expand to reload the damper, allowing the crutch to return to its initial length in preparation for the next applied force. This will be later elaborated in the Spring-Damper Optimization section of this report. The spring-damper system is shown in the figure below:

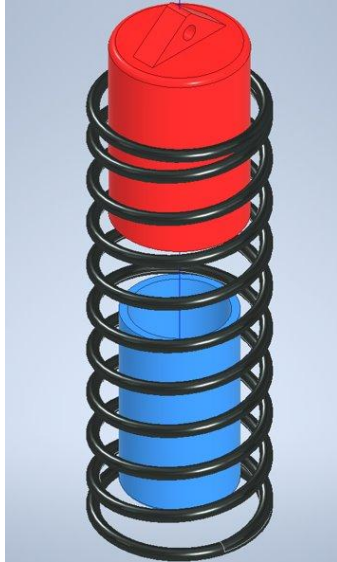


Figure 22: Mechanical Spring-Damper Assembly

ADJUSTABLE PRELOAD MECHANISM

It was critical for our design to have an adjustable preload in order for optimal performance by various users. The spring stiffness of the crutch should be selected to receive maximum force dissipation from the damper. For this reason, the spring needs to have an adjustable spring rate. The spring rate of a spring refers to the amount of force required to compress or extend the spring by a given amount. Users of different body mass will each exert a different force on the spring, which will change the force dissipation of the damper. For this reason, a mechanism is required to manually adjust the preload of the spring. This mechanism will only need to be actuated once per user, so it does not need to be easily adjustable.

The mechanism our group designed includes two threaded hexagonal nuts that can be used to exert an upwards force on the spring and create a given compression. This can be seen in the figure below.

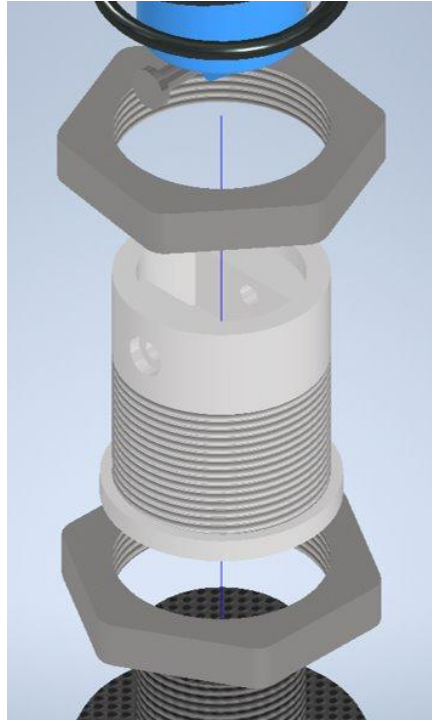


Figure 23: Adjustable Preload Mechanism

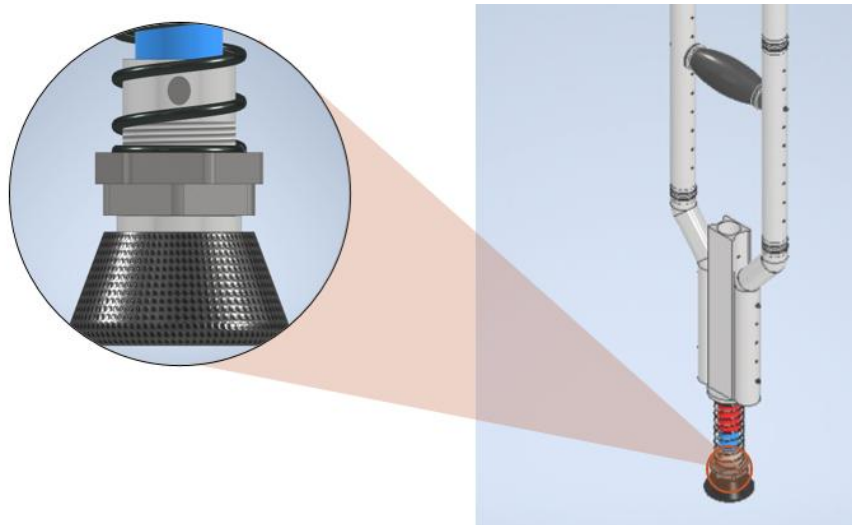


Figure 24: Adjustable Preload Mechanism Location

This mechanism will be included at the bottom of the crutch beneath the spring to allow for the spring rate to be manually adjusted to its optimal value.

ADJUSTABLE BASES

Many different bases were created in CAD that can be easily threaded onto the base of the crutch. Standard axillary crutches include only one base, and this base is designed mostly for indoor flooring. In the event that the user is traveling outdoors or on abnormal floor surfaces, we created designs of various materials and dimensions to allow for different contact areas. Four of the designed bases can be seen in the figure below.

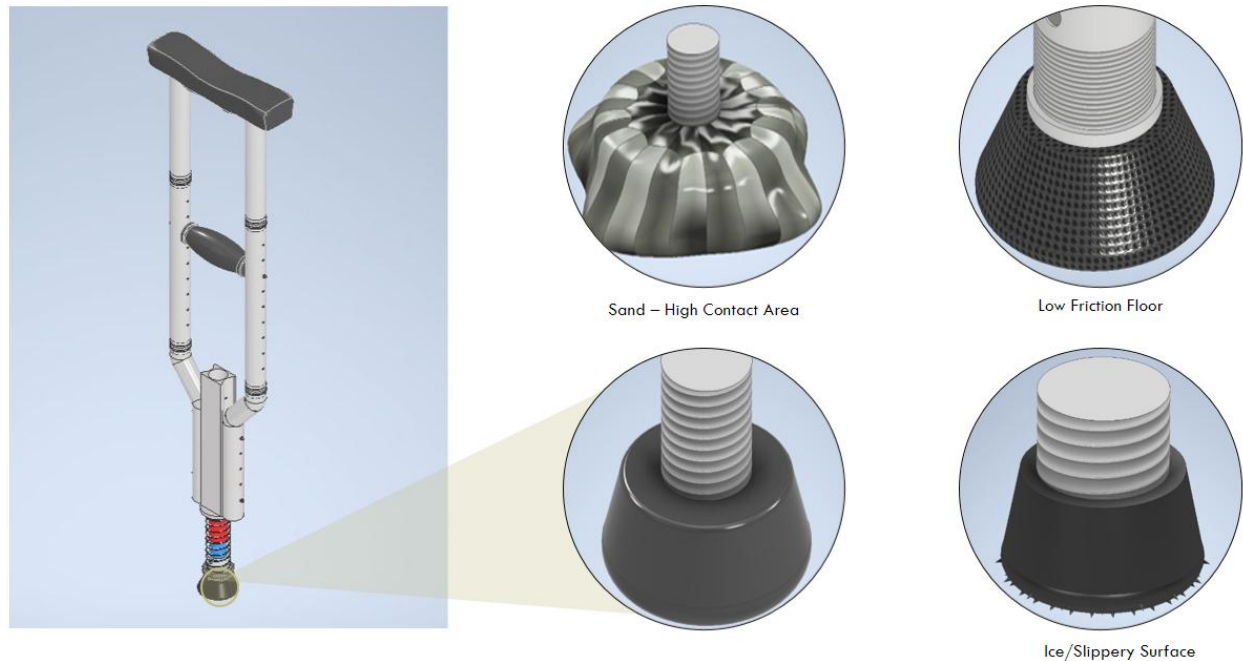


Figure 25: Various Bases

Spring-Damper Optimization

In order to optimize the parameters used in the spring-damper system, a LaGrangian analysis was conducted to determine the kinematic motion of the crutch. The analysis was conducted in two phases: the loading phase and the swing phase. In the loading phase, the crutch is rotating and bearing the weight of the subject. In the swing phase, the crutch is simply lifted off of the ground, with no rotational motion. Based on literature sources, the stance and swing phase were assumed to both last 0.75 seconds each [14]. Initial conditions for the system was also chosen with reference to a study conducted by Liu et. Al [15]. A system of differential equations was constructed in MATLAB for each loading phase and were solved using ODE45.

In order to choose a minimum spring constant, the radial displacement of the crutch was plotted over time. A minimum spring constant would allow for the length of the crutch **and the end of the swing phase** to match the original length of the crutch **at the beginning of the loading phase**; the crutch must re-expand back to its original length before it is compressed again. Based on these conditions, a spring constant of 22000 N/m was found to be sufficient. Figure 26 shows the plot of the radial displacement of the crutch for a variety of spring constants.

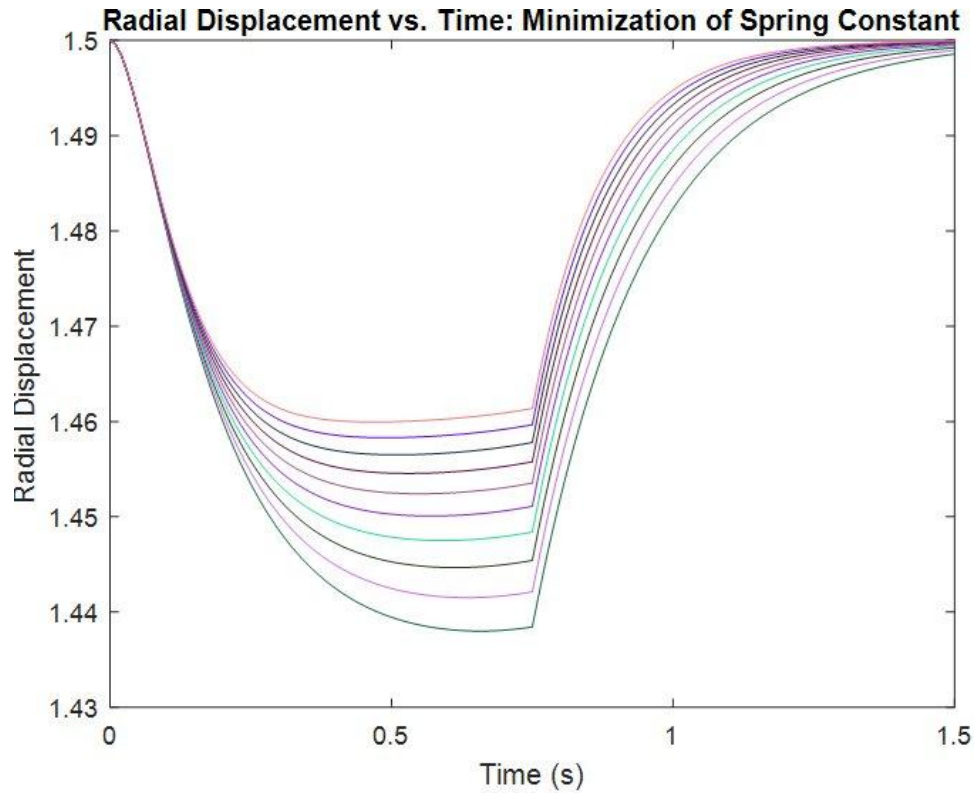


Figure 26: Radial displacement of crutch over time. As spring constant increases, vertical position of function increases as well

In order to choose a minimum damping constant, the spring reaction force was plotted over time. A minimum damping coefficient would ensure that the spring force **does not** oscillate and that the maximum force is below the body weight. Based on these conditions, a damping constant of 2600 Ns/m was found to be sufficient. Figure 27 shows the plot of spring force over time for a variety of damping coefficients.

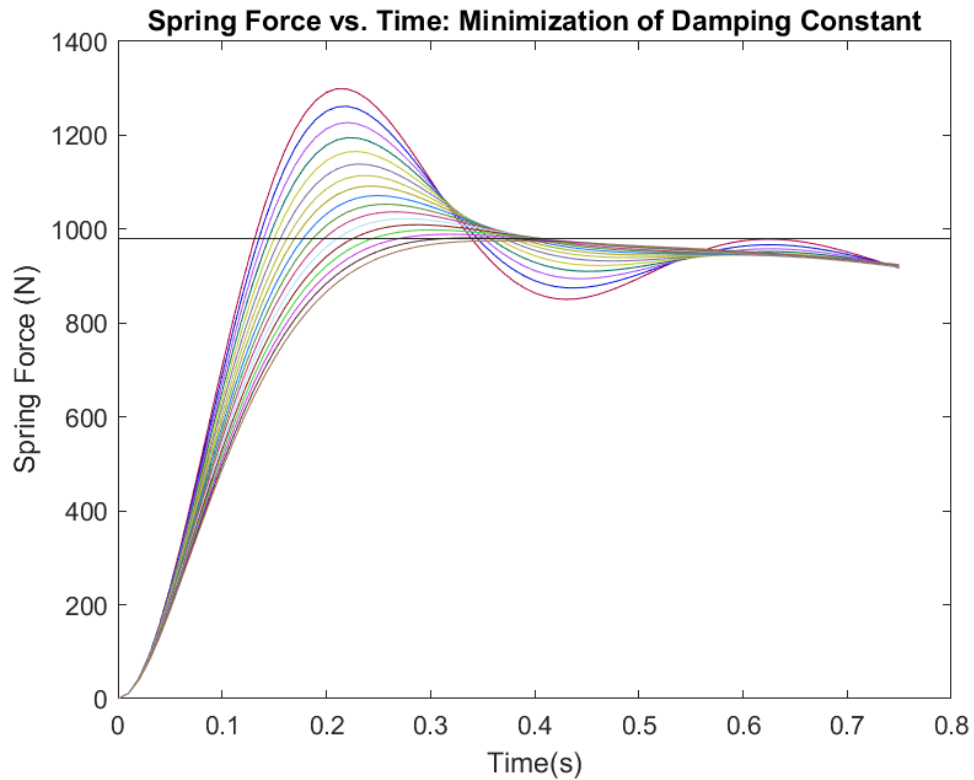


Figure 27: Spring force over time. As damping constant increases, maximum force decreases

The effects of increasing the damping coefficient past the minimum value were also investigated and plotted in Figure 28. The red line shows the reaction force applied under the arm if no spring-damper system were used. All other lines show the reaction force decreasing as the damping coefficient increases. Based on this plot it can also be seen that the maximum reaction force **with the damping system** is always less than the maximum force without. As well, the reaction force with the damping system increases over time and the area under the curve (impulse) is smaller than without. Therefore, it can be concluded that our objectives to reduce the maximum force and impulse were accomplished.

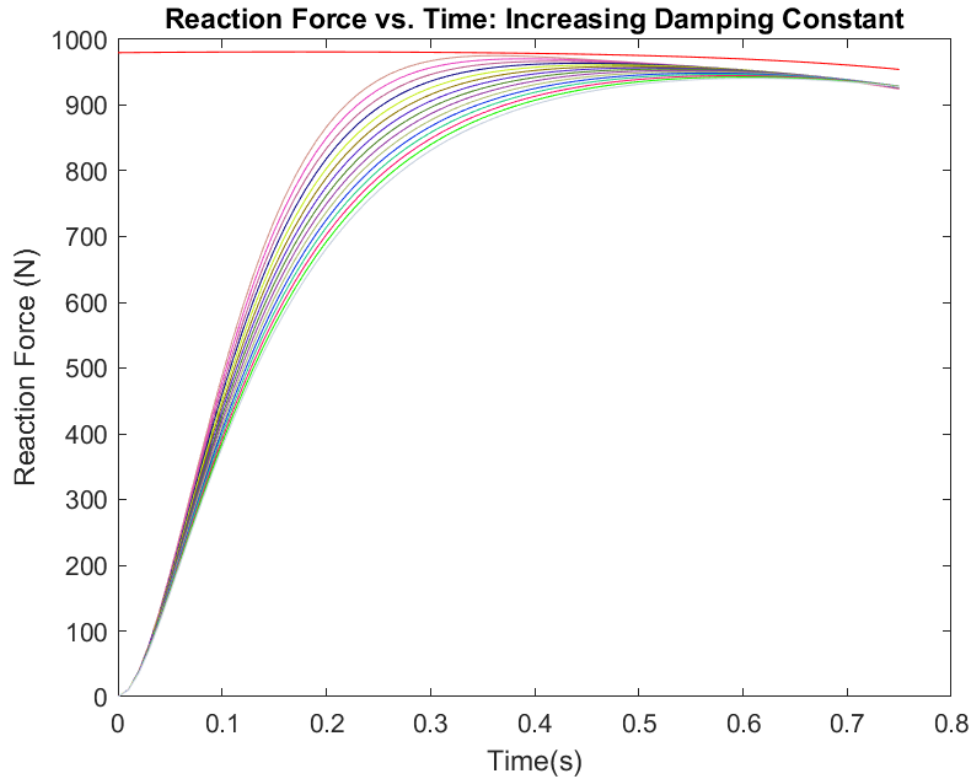


Figure 28: Reaction force over time. As damping constant increases, maximum force decreases. Red line indicates reaction force in absence of spring-damper system

Details on the LaGrangian analysis conducted are in *Appendix A*. MATLAB code used and optimization procedures are in *Appendix B*.

Mathematical Stress Analysis

An overall factor of safety of 5 was used to assess the safety of our design. Current manufacturing standards for medical devices have a minimum safety factor of 4 for brittle parts, and 2.5 otherwise [16]. 4 critical locations were identified, and the system was also assessed for column buckling. Overall, the modes of failure were defined as follows:

1. The curved section of the beam that is loaded in bending
2. The transverse pin in the handhold and height adjustment that are in transverse shear
3. The adjustment holes that are in bending and introduce regions of stress concentration
4. The threaded portion of the lock that is in axial loading

These locations can be visualized in Figure 29.

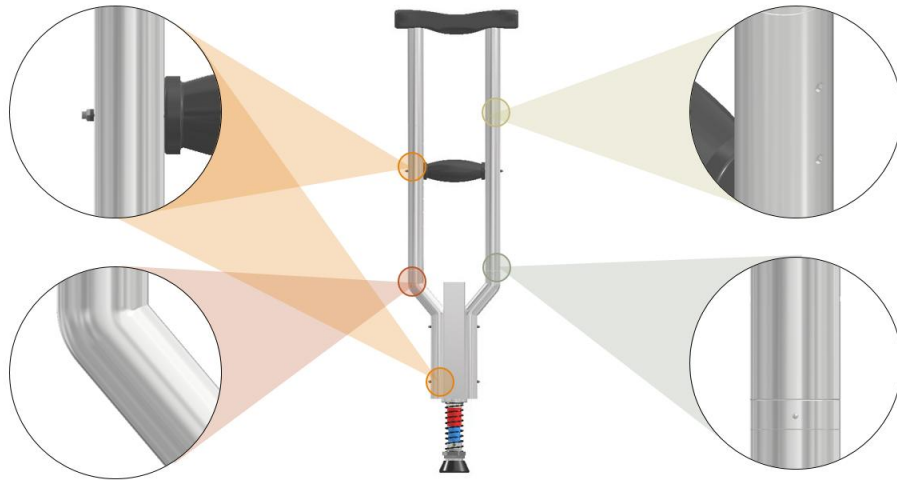


Figure 29: Critical points in crutch design

The safety factors were calculated for each of the critical points in the crutch design. These calculations were completed based on the concepts taught in MECHENG 3E05 and with assumed knowledge of previous classes. In order to complete these calculations, several simplifying assumptions for the structure of the crutch and the loading scenario had to be made. These general assumptions are summarized below, and assumptions specific to a critical point are found within that respective section of the hand calculations.

1. The system is in static equilibrium
2. Both crutches are equally supporting the body weight (therefore the force applied to each crutch is half of the body weight)
3. All poles of the crutch are assumed to be rigid bodies
4. Analysis is taking place at the point where the crutch is exactly perpendicular to the ground
5. Crutch is cyclically loaded at each critical point
6. The minimum stress is zero
7. The locking mechanism is not load bearing
8. No torsional or transverse stresses exist as no torque is applied during loading (assuming the user operates the crutch appropriately)

It is important to note that each of the safety factors calculated represent **fatigue factors** rather than just static loading. Goodman's failure locus was used to find these factors of safety in fatigue. In addition, the number of cycles to failure was calculated to ensure that the lifetime of the device is considered infinite.

Appendix C includes hand calculations for the factor of safety and the specific solution procedure for each critical point. In addition, a MATLAB code was created to validate our hand calculations, which is provided in *Appendix D*. This helped us to determine the appropriate crutch dimensions and material properties to obtain acceptable safety factors that were greater than 5. *Table 3* summarizes the factors of safety for a variety of aluminum alloys at different ultimate tensile strengths.

Table 3: Factors of safety for each critical point with increasing ultimate tensile strength

Ultimate Strength	Factor of Safety
-------------------	------------------

	Curved Beam in Bending	Pin Shear	Threaded Beam in Axial Loading	Transverse Holes in Bending
324 MPa (2011 T6-Al Alloy)	6.49	6.22	4.62	2.80
446 MPa (2024 T6-Al Alloy)	8.93	8.56	6.36	3.85
593 MPa (7075 T6-Al Alloy)	11.88	11.39	8.46	5.12

Based on the calculations for factors of safety in fatigue, the most critical loading points of the crutch are the holes in the pole of the crutch that allow the handhold to be adjustable. Stress concentrates close to these holes and as a result the factor of safety is the lowest. Therefore, for all elements to have safety factor over 5 as discussed previously, the ultimate tensile strength of the material must be 593 MPa, corresponding to the 7075 T6 aluminum alloy. Therefore, we are able to find a suitable material that produces an acceptable factor of safety for each critical point in the crutch at the corresponding dimensions. With this specific aluminum alloy used for the crutch, we confirmed through the stress analysis that our crutch design will be able to withstand regular loading by the user without failing as the crutch has an infinite life.

Cost Analysis

Table 4: Cost analysis for the manufacturing of one set of Bound-Less Crutches

Item	Items/Unit	Cost/Item	Total Cost
Adjustable shock absorbers	2.00	115.85	231.70
8 ft. 7075-T6 aluminum pipe	2.00	48.52	97.04
M5 screws for handle and height adjustment	6.00	0.44	2.63
Compression springs for joints	4.00	4.79	19.16
High density foam	1.00	0.97	0.97
Elastic rope	0.02	17.00	0.38
Manufacturing cost estimate (CNC shop \$/hour)	0.50	88.31	44.16
Total Price of One Set of Crutches*			396.04

*All prices in CAD

*Assuming 100 sets of crutches are made

As per Table 4 above, the cost to manufacture our product is roughly \$400. These costs were accounted for in Canadian dollars, though many of our parts and materials were sourced in the United States. For the

purpose of considering bulk discounts, we have assumed that 100 pairs of crutches will be made. It is impossible to say what the true market potential is of our product without further surveying and research so this assumption was adopted as a worst-case scenario. Where our exact specification of a product is not available to purchase as a standard option, as such is the case for our aluminum tubing, an average of costs for similar dimensions of material was used.

It is clear that this product cannot be a high volume, budget solution to the problem that crutches address. Though, from the scope of our design, it was never intended to be. This justifies the expense of components like aluminum tubing and adjustable shock absorbers. However, it means that any bulk discounts beyond what is accounted for above will likely not be significant. It also means that our manufacturing cost will be quite high since it could require the skilled application of several manufacturing practices for a relatively insignificant amount of time. Further, any tooling, cnc programming, 3D printing, or other fabrication processes will come at a premium cost per item.

While the cost of the initial prototype will be high, there are undoubtedly steps we can take to reduce the cost of the final product before pronging it to market. We have outlined a few of these below as a portion of the next stages of the design process for Bound-Less Crutches.

Next Steps and Conclusions

With the goal of bringing Bond-Less Crutches to market, our design must undergo further stages of development. The first of these would be prototyping, either creating a fully functional set or a series of testing rigs to address the following:

- Determining the effectiveness, useability, and durability of our joint locking mechanism geometry as well as the hardware it contains.
- Assessing our handles and under-shoulder pads through group testing to ensure they are comfortable and supportive for a wide range of users.
- Evaluating our use of shock absorbers, both to confirm that they provide the desired force-reducing effect, and to make sure the additional mass toward the end of the crutch does not make it unwieldy.
- Using our crutches in a variety of typical scenarios such as sitting down, stairs, and inclines to make sure they perform equal-to if not better-than traditional axillary crutches regarding comfort and usability.
- Observing failure modes during improper and unusual use of the crutch to minimize the risk of injury in all scenarios.

These targeted assessments, as well as more general testing, would allow us to iterate on this initial design and likely achieve an improved product which we would be confident to introduce to the public.

Additionally, it will benefit us to improve our understanding of the kinematics of our design and its user. We would like to expand on the LaGrangian analysis by removing simplifying assumptions. Currently, our analysis assumes that the weight of the body is centralized at the top of the crutch. An expansion would be accounting for the toe-off force and torque applied to the system at the beginning of the loading phase and the changing centre of mass of the body.

As mentioned above, there is also scope for refinement through cost reduction. The first item we would seek to address is the aluminum tubing which forms the structure of the crutch. Through testing, we may determine that there is some budget in our scope to add more weight and still hit the design objectives. This would allow us to use a less expensive metal such as steel. Alternatively, we may be able to alter the

geometry of our tubing such that a hard plastic would still meet our load bearing safety requirements. Either alternative would reduce our material cost significantly.

Another avenue we could pursue with the goal of cost reduction would be designing our own shock absorbers. Our off-the-shelf solution is by far the biggest expense of our design. While they meet our needs in being small, adjustable, and having a low rebound force, they are rated to support 500 lbs, which is more than double our required spec. If we were to design our own dashpot and housing assembly using the ideal parameters calculated in our spring-damper optimization, we could likely not only reduce the cost of this item, but also its complexity and weight. This would pay dividends in the useability of the crutches as well resulting in a much-improved overall product.

Finally, the clinical efficacy of our design should be validated in a randomized trial or series of case studies to confirm that the incidence of nerve-palsy decreases as a result of usage.

Our design was able to sufficiently meet our design objectives as follows:

- Accommodate a range of body shapes due to the adjustable handgrips and height
- Accommodate a range of body sizes due to its geometry, material selection, and stress-analysis conducted for the weight of 95th percentile males
- Fold into a portable and compact form that
 - Is one-handed to ensure intuitive and easy use for users with limited mobility
 - Is fast to assemble and disassemble due to a maximum of 3 assembly steps
- Intuitive and easy for users with limited mobility
- Fast to assemble and disassemble (under 1 minute)
- Reduce maximum force and impulse applied under arm due to spring damper system as verified by LaGrangian analysis
- Utilize an aluminum alloy that satisfies safety criteria, is lightweight, and is durable.

Overall, our project was able to successfully meet our initial objectives and has the potential to be expanded in the future.

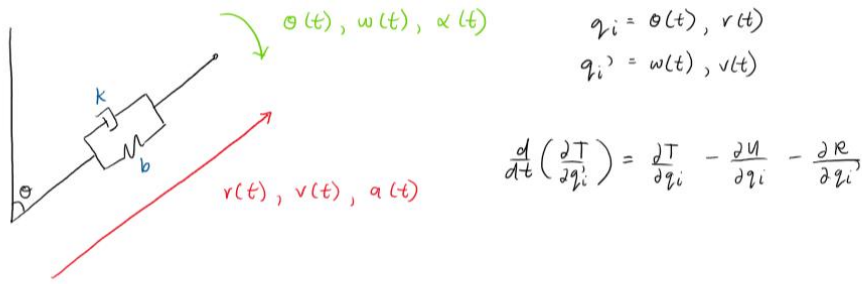
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Appendix A: Lagrangian Analysis



$$T = \text{kinetic energy} = \frac{1}{2} m v(t)^2 + \frac{1}{2} m (r(t) \omega(t))^2$$

$$= \frac{1}{2} m v(t)^2 + \frac{1}{2} m r(t)^2 \omega(t)^2$$

$$U = \text{potential energy} = m g r(t) \cos(\theta(t)) + \frac{1}{2} k (L - r(t))^2$$

$$R = \text{energy dissipated} = \frac{1}{2} b v(t)^2$$

$$\textcircled{1} q_i = \theta(t)$$

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\omega}} \right) = \frac{\partial T}{\partial \theta} - \frac{\partial U}{\partial \theta} - \frac{\partial R}{\partial \dot{\omega}}$$

$$\frac{\partial T}{\partial \dot{\omega}} = \frac{\partial}{\partial \dot{\omega}} \left(\frac{1}{2} m v(t)^2 + \frac{1}{2} m r(t)^2 \omega(t)^2 \right)$$

$$= \frac{\partial}{\partial \dot{\omega}} m r(t)^2 \omega(t)$$

$$= m r(t)^2 \omega(t)$$

$$\frac{d}{dt} (m r(t)^2 \omega(t)) = m \omega(t) \cdot \frac{d}{dt} (r(t)^2) + m r(t)^2 \cdot \frac{d}{dt} (\omega(t))$$

$$= m \omega(t) (2 r(t) v(t)) + m r(t)^2 \alpha(t)$$

$$= 2 m r(t) v(t) \omega(t) + m r(t)^2 \alpha(t)$$

$$\frac{\partial T}{\partial \theta} = \frac{\partial}{\partial \theta} \left(\frac{1}{2} m v(t)^2 + \frac{1}{2} m r(t)^2 \omega(t)^2 \right) = 0$$

$$\frac{\partial U}{\partial \theta} = \frac{\partial}{\partial \theta} (m g r(t) \cos(\theta(t))) = -m g r(t) \sin(\theta(t))$$

$$\frac{\partial R}{\partial \dot{\omega}} = \frac{\partial}{\partial \dot{\omega}} \left(\frac{1}{2} b v(t)^2 \right) = 0$$

overall Lagrangian along θ :

$$2 m r(t) v(t) \omega(t) + m r(t)^2 \alpha(t) = m g r(t) \sin(\theta(t))$$

$$2 v(t) \omega(t) + r(t) \alpha(t) = g \sin(\theta(t))$$

$$\alpha(t) = \frac{g \sin(\theta(t)) - 2 v(t) \omega(t)}{r(t)}$$

$$② \quad q_i = r(t)$$

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{v}} \right) = \frac{\partial T}{\partial r} - \frac{\partial U}{\partial r} - \frac{\partial R}{\partial v}$$

$$\begin{aligned} \frac{\partial T}{\partial v} &= \frac{\partial}{\partial v} \left(\frac{1}{2} m v(t)^2 + \frac{1}{2} m r(t)^2 \omega(t)^2 \right) \\ &= \frac{2}{2} m v(t) = m v(t) \end{aligned}$$

$$\frac{d}{dt} (m v(t)) = m a(t)$$

$$\begin{aligned} \frac{\partial T}{\partial r} &= \frac{\partial}{\partial r} \left(\frac{1}{2} m v(t)^2 + \frac{1}{2} m r(t)^2 \omega(t)^2 \right) \\ &= \frac{2}{2} m r(t) \omega(t)^2 \\ &= m r(t) \omega(t)^2 \end{aligned}$$

$$\begin{aligned} \frac{\partial U}{\partial r} &= \frac{\partial}{\partial r} (m g r(t) \cos(\theta(t)) + \frac{1}{2} k (L - r(t))^2) \\ &= m g \cos(\theta(t)) + \frac{2}{2} (-1) k (L - r(t)) \\ &= m g \cos(\theta(t)) - k (L - r(t)) \end{aligned}$$

$$\begin{aligned} \frac{\partial R}{\partial v} &= \frac{\partial}{\partial v} \left(\frac{1}{2} b v(t)^2 \right) = \frac{2}{2} b v(t) \\ &= b v(t) \end{aligned}$$

overall Lagrangian:

$$m a(t) = m r(t) \omega(t)^2 - m g \cos(\theta(t)) + k (L - r(t)) - b v(t)$$

$$a(t) = r(t) \omega(t)^2 - g \cos(\theta(t)) + \frac{k}{m} (L - r(t)) - \frac{b}{m} v(t)$$

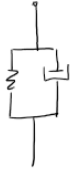
Overall system of equations:

$$\frac{dr}{dt} = v$$

$$\frac{d^2 r}{dt^2} = r \omega^2 - g \cos(\theta) + \frac{k}{m} (L - r) - \frac{b}{m} v$$

$$\frac{d\theta}{dt} = \omega$$

$$\frac{d^2 \theta}{dt^2} = \frac{g \sin(\theta) - 2v\omega}{r}$$



$r(t), v(t), a(t)$

ONLY consider mass of crutch / bottom

No angular motion ($\theta(t), \omega(t), \alpha(t)$)

↳ crutch is vertical @ 90°

$m = 1 \text{ kg}$

Kinetic energy of system: $T = \frac{1}{2} m v(t)^2$

Potential energy of system: $U = \frac{1}{2} k (L - r(t))^2$

Energy dissipated in system: $R = \frac{1}{2} b v(t)^2$

$$q_i = r(t) \longrightarrow \frac{d}{dt} \left(\frac{\partial T}{\partial v} \right) = \frac{\partial T}{\partial r} - \frac{\partial U}{\partial r} - \frac{\partial R}{\partial v}$$

$$\frac{\partial T}{\partial v} = m v(t) \quad \frac{d}{dt} (m v(t)) = m a(t)$$

$$\frac{\partial T}{\partial r} = 0 \quad \frac{\partial U}{\partial r} = -k (L - r(t)) \quad \frac{\partial R}{\partial v} = b v(t)$$

$$m a(t) = k (L - r(t)) - b v(t)$$

$$a(t) = \frac{k (L - r(t)) - b v(t)}{m}$$

overall system of equations:

$$\frac{dr}{dt} = v(t) \quad \frac{d^2 r}{dt^2} = \frac{k (L - r(t)) - b v(t)}{m_d} \quad m_d = 0.5 \text{ kg}$$

Overall system of equations:

$$\frac{dr}{dt} = v$$

$$\frac{d^2 r}{dt^2} = \frac{k(L - r) - bv}{m_d}$$

Appendix B: MATLAB Code for Spring-Damper Optimization

```
% Optimization
```

```
clear all;
```

```
close all;
```

```
clc;
```

```
% Variables
```

```
global l m g k b md
```

```
l = 1.5; %Crutch length, m
```

```
m = 100; %Mass of user, kg
```

```
g = 9.81; %Gravitational acceleration, m^2/s
```

```
b = 3000; %Damping coefficient, Ns/m
```

```
md = 0.5; %Mass of damper system
```

```

% Optimize the spring coefficient
cond = 0;

for k = 15000:1000:40000

    % Loading phase
    % Initial conditions
    r_0 = 1; %Initial radius is equal to length of crutch
    v_0 = 0; %Initial radial velocity
    theta_0 = -0.05; %12 degrees
    omega_0 = 0.30; %Initial tangential velocity from literature
    L_0 = [r_0, v_0, theta_0, omega_0];

    % Time span
    t_span1 = [0:0.01:0.75];

    % Solve system
    [t,L] = ode45(@Lagrangian, t_span1, L_0);

    % Lift-off phase
    % Initial conditions
    r_0 = L(76,1);
    v_0 = L(76,2);
    L_0 = [r_0, v_0];

    % Time span
    t_span2 = [0.75:0.01:1.5];

    % Call function/solve ODE
    [te,Le] = ode45(@Expand, t_span2, L_0);

    % Set line colour
    col = rand(1,3);

    % Plot radial displacement over both cycles
    name = string(k);
    figure(1)
    plot(t, L(:,1), 'color', col)
    hold on
    plot(te, Le(:,1), 'color', col)
    xlabel('Time (s)')
    ylabel('Radial Displacement')
    title('Radial Displacement vs. Time: Minimization of Spring Constant')

    cond = Le(76, 1);

    if cond > (1-0.0001)

        k_chosen = k;
        fprintf("The optimal k value is %d \n", k_chosen)
        break

    end

end
end

```

```

%Optimize the damping coefficient
max = 1000;
for b = 1000:100:4000

    % Loading phase
    % Initial conditions
    r_0 = 1; %Initial radius is equal to length of crutch
    v_0 = 0; %Initial radial velocity
    theta_0 = -0.05; %12 degrees
    omega_0 = 0.30; %Initial tangential velocity from literature
    L_0 = [r_0, v_0, theta_0, omega_0];

    % Time span
    t_span1 = [0:0.01:0.75];

    % Solve system
    [t,L] = ode45(@Lagrangian, t_span1, L_0);

    % Calculate spring force
    Fs = k*(1-L(:,1));

    figure(2)
    col = rand(1,3);
    plot(t, Fs, 'color', col)
    hold on
    yline(m*g)
    xlabel('Time(s)')
    ylabel('Spring Force (N)')
    title('Radial Displacement vs. Time: Minimization of Damping Constant')

    max = findpeaks(Fs);

    if max < m*g
        b_min = b;
        fprintf('The minimum damping coefficient is %d \n', b_min)

        break
    end
end

% Determine reaction force without spring damper system
Fn = m*g*cos(L(:,3));

% Determine effect of increasing b-value past minimum
for b = b_min:100:4000

    % Loading phase
    % Initial conditions
    r_0 = 1; %Initial radius is equal to length of crutch
    v_0 = 0; %Initial radial velocity
    theta_0 = -0.05; %12 degrees
    omega_0 = 0.30; %Initial tangential velocity from literature
    L_0 = [r_0, v_0, theta_0, omega_0];

```

```

% Time span
t_span1 = [0:0.01:0.75];

% Solve system
[t,L] = ode45(@Lagrangian, t_span1, L_0);

% Calculate spring force
Fs = k*(1-L(:,1));

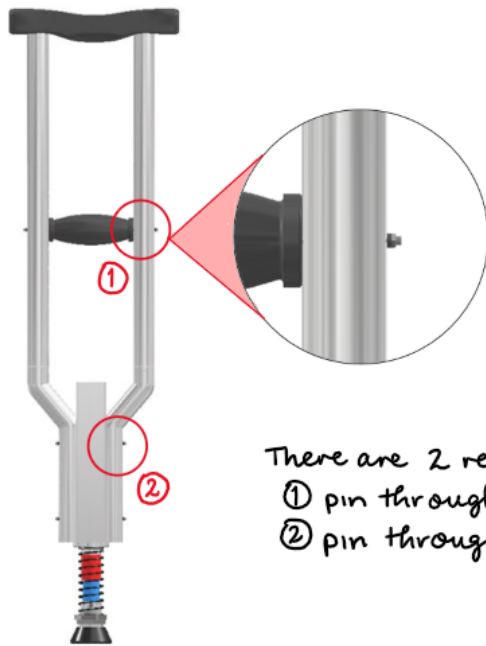
figure(3)
col = rand(1,3);
plot(t, Fs, 'color', col)
hold on
plot(t, Fn, 'r-')
xlabel('Time(s)')
ylabel('Reaction Force (N)')
title('Increasing b-Values past minimum')

end

```

Appendix C: Stress Analysis Hand Calculations

1. Pin Shear



There are 2 regions of interest for pin shear:
 ① pin through handle
 ② pin through base of crutch

Assumptions:

- pins are loaded in transverse shear only, not in bending

Known Values & Dimensions:

- $D = 5 \times 10^{-3} \text{ m}$ (diameter of pin)

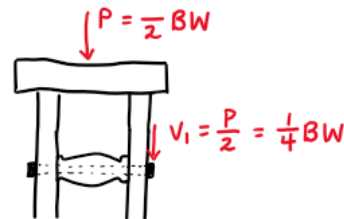
- $BW = 1000 \text{ N}$ (body weight force on crutch)

For Region ①:

↳ this is single shear

$$\tau_{\max,1} = \frac{V_1}{A} = \frac{\frac{1}{4}BW}{\frac{\pi}{4}D^2} = \frac{\frac{1}{4}(1000 \text{ N})}{\frac{\pi}{4}(5 \times 10^{-3} \text{ m})^2}$$

$$\tau_{\max,1} = 1.273 \times 10^7 \text{ Pa} = 12.73 \text{ MPa}$$

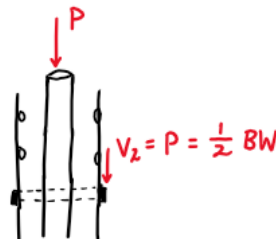


For Region ②:

↳ this is double shear

$$\tau_{\max,2} = \frac{V_2}{2A} = \frac{\frac{1}{2}BW}{2 \cdot \frac{\pi}{4}D^2} = \frac{\frac{1}{2}(1000 \text{ N})}{2 \cdot \frac{\pi}{4}(5 \times 10^{-3})^2}$$

$$\tau_{\max,2} = 1.273 \times 10^7 \text{ Pa} = 12.73 \text{ MPa}$$



Note: $\tau_{\max,1} = \tau_{\max,2}$

In this case, pin shear is the only source of stress

$$\hookrightarrow \sigma_a = T_{\max}/2 = 6.365 \text{ MPa}$$

$$\sigma_m = T_{\max}/2 = 6.365 \text{ MPa}$$

$S_{ut} = 593 \text{ MPa}$ (ultimate strength in tension for 7075 Aluminum Alloy T6)

$S_{uts} = S_{ut}/2 = 296.5 \text{ MPa}$ (ultimate strength in shear)

Find S_e , where $S_e = k_a k_b k_c k_d k_e k_f S_e'$

$$S_e' = 0.5 S_{uts} = 148.25 \text{ MPa}$$

$$k_a = 1$$

$$k_b = 1.24 D^{-0.107} = 1 \text{ (for } 2.79 < D < 51 \text{ mm)}$$

$$k_c = 1$$

$$k_d = 1$$

$$k_e = 0.62 \text{ (use highest reliability of 99.9999\%)}$$

$$k_f = 1$$

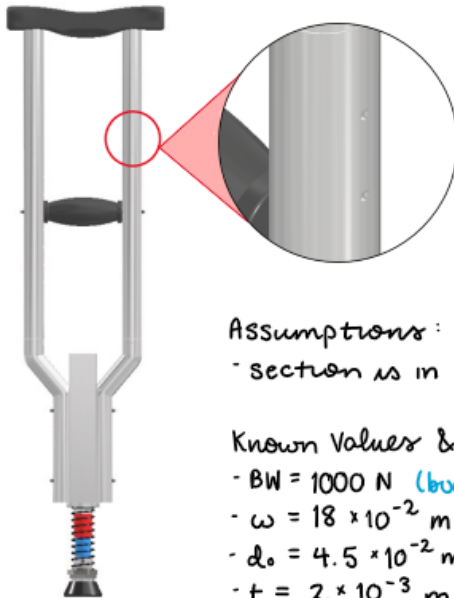
$$S_e = (1)(1)(1)(1)(0.62)(1)(148.25 \text{ MPa}) = 91.92 \text{ MPa}$$

Find safety factor: $n_f = \left(\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} \right)^{-1}$ (using Goodman's criteria)

$$n_f = \left(\frac{6.365}{91.92} + \frac{6.365}{296.5} \right)^{-1}$$

$$n_f = 11.02$$

2. Transverse Holes in Bending

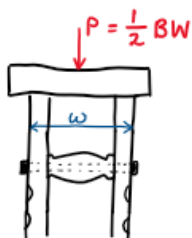


Assumptions :

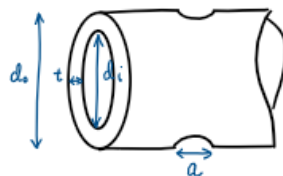
- section is in bending only

Known Values & Dimensions:

- $BW = 1000 \text{ N}$ (body weight force on crutch)
- $w = 18 \times 10^{-2} \text{ m}$ (width of crutch)
- $d_o = 4.5 \times 10^{-2} \text{ m}$ (outer diameter of crutch)
- $t = 2 \times 10^{-3} \text{ m}$ (thickness of crutch wall)
- $a = 5 \times 10^{-3} \text{ m}$ (diameter of transverse hole)
- $d_i = d_o - 2t = 4.1 \times 10^{-2} \text{ m}$ (inner diameter of crutch)



cross-sectional view of crutch pole :



Calculate moment from P @ outer surface of crutch:

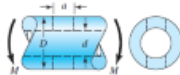
$$M = \frac{1}{2} w \cdot P = \frac{1}{2} w \cdot \frac{1}{2} BW = \frac{1}{4} (1000 \text{ N})(18 \times 10^{-2} \text{ m}) = 45 \text{ Nm}$$

From Shigley's Mechanical Engineering Textbook :

Table A-16

Approximate Stress-Concentration Factor K_t of a Round Bar or Tube with a Transverse Round Hole and Loaded in Bending

Source: R. E. Peterson, Stress-Concentration Factors, Wiley, New York, 1974, pp. 146, 235.



The nominal bending stress is $\sigma_b = M/Z_{net}$ where Z_{net} is a reduced value of the section modulus and is defined by

$$Z_{net} = \frac{\pi A}{32D} (D^3 - d^3)$$

Values of A are listed in the table. Use $d = 0$ for a solid bar

a/D	0.9		d/D 0.6		0	
	A	K_t	A	K_t	A	K_t
0.050	0.92	2.63	0.91	2.55	0.88	2.42
0.075	0.89	2.55	0.88	2.43	0.86	2.35
0.10	0.86	2.49	0.85	2.36	0.83	2.27
0.125	0.82	2.41	0.82	2.32	0.80	2.20
0.15	0.79	2.39	0.79	2.29	0.76	2.15
0.175	0.76	2.38	0.75	2.26	0.72	2.10
0.20	0.73	2.39	0.72	2.23	0.68	2.07
0.225	0.69	2.40	0.68	2.21	0.65	2.04
0.25	0.67	2.42	0.64	2.18	0.61	2.00
0.275	0.66	2.48	0.61	2.16	0.58	1.97
0.30	0.64	2.52	0.58	2.14	0.54	1.94

Nominal bending stress: $\sigma_o = \frac{M}{Z_{net}}$

$$Z_{net} = \frac{\pi A}{32D} (D^4 - d^4) \rightarrow \text{find } A \text{ from Table A-16}$$

$$\rightarrow \text{for } \frac{a}{D} = \frac{5 \times 10^{-3}}{4.5 \times 10^{-2}} = 0.111, \frac{d}{D} = \frac{4.1 \times 10^{-2}}{4.5 \times 10^{-2}} = 0.911:$$

-assume $d/D \sim 0.9$, interpolate between $a/D = 0.1$ & 0.125

$$\frac{0.11 - 0.1}{0.125 - 0.1} = \frac{A - 0.86}{0.82 - 0.86} \Rightarrow A = 0.844$$

$$Z_{net} = \frac{\pi(0.844)}{32(4.5 \times 10^{-2})} [(4.5 \times 10^{-2})^4 - (4.1 \times 10^{-2})^4] = 2.3474 \times 10^{-6} \text{ m}^3$$

$$\sigma_o = \frac{M}{Z_{net}} = \frac{45 \text{ Nm}}{2.3474 \times 10^{-6} \text{ m}^3} = 1.917 \times 10^7 \text{ Pa} = 19.17 \text{ MPa}$$

Find K_t from Table A-16: (interpolate again)

$$\frac{0.11 - 0.1}{0.125 - 0.1} = \frac{K_t - 2.49}{2.41 - 2.49} \Rightarrow K_t = 2.458$$

Calculate stress mean & amplitude:

$$\sigma_a = \frac{1}{2} K_t \sigma_o = \frac{1}{2} (2.458) (19.17 \text{ MPa}) = 23.56 \text{ MPa}$$

$$\sigma_m = \frac{1}{2} K_t \sigma_o = \frac{1}{2} (2.458) (19.17 \text{ MPa}) = 23.56 \text{ MPa}$$

$S_{ut} = 593 \text{ MPa}$ (ultimate strength in tension for 7075 Aluminum Alloy T6)

Find S_e , where $S_e = k_a k_b k_c k_d k_e k_f S_e'$

$$S_e' = 0.5 S_{ut} = 296.5 \text{ MPa}$$

$$k_a = 1$$

$$k_b = 1.24 d_o^{-0.107} = 0.825 \text{ (for } 2.79 < d_o < 51 \text{ mm)}$$

$$k_c = 1$$

$$k_d = 1$$

$$k_e = 0.62 \text{ (use highest reliability of 99.999\%)}$$

$$k_f = 1$$

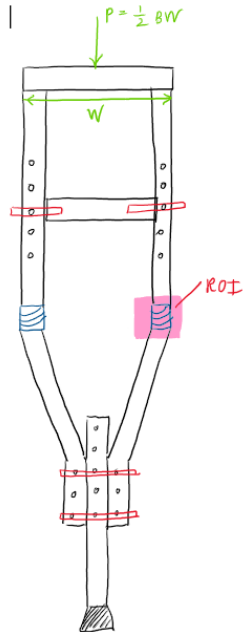
$$S_e = (1)(0.825)(1)(1)(0.62)(1)(296.5 \text{ MPa}) = 151.66 \text{ MPa}$$

Find safety factor: $n_f = \left(\frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_{ut}} \right)^{-1}$ (using Goodman's criteria)

$$n_f = \left(\frac{23.56}{151.66} + \frac{23.56}{593} \right)^{-1}$$

$$n_f = 5.12$$

3. Threaded Beam in Axial Loading



Assumptions:

- threads are under axial loading, $V = P/2$
- pitch is 2mm, there are 4 turns (n_t)

① calculate d_r

$$d_r = 1.226869 p \quad p = 2 \text{ mm}$$

$$= (1.226869)(2 \text{ mm}) = 2.4537 \text{ mm}$$

② calculate τ_{axial}

$$\tau_{\text{axial}} = \frac{-4V}{\pi d_r^2} \quad V = \frac{P}{2} = \frac{1}{4} BW = 250 \text{ N}$$

$$\tau_{\text{axial}} = \frac{-4(250 \text{ N})}{\pi (2.4537 \times 10^{-3})^2} = -52.869 \text{ MPa}$$

③ calculate τ_{bending}

$$\tau_{\text{bending}} = \frac{6V}{\pi d_r n_t p} \quad n_t = 4$$

$$\tau_{\text{bending}} = \frac{6(250 \text{ N})}{\pi (2.4537 \times 10^{-3})(4)(2 \times 10^{-3})} = 24.323 \text{ MPa}$$

④ calculate total stress

$$\tau_{\text{max}} = \tau_{\text{axial}} + \tau_{\text{bending}}$$

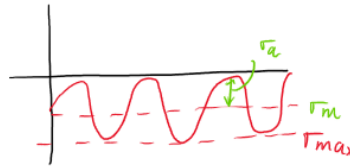
$$= -52.869 + 24.323 = -28.546 \text{ MPa}$$

$$\tau_a = \tau_m = |\tau_{\text{max}}/2| = 14.273 \text{ MPa}$$

⑤ calculate endurance limit

$$S_e' = 0.5 (S_{ut}) = 0.5 (593 \text{ MPa})$$

$$= 296.5 \text{ MPa}$$



$$K_a = 1$$

$$K_b = 1.24 (4s)^{-0.107} = 0.825$$

$$K_c = K_d = K_f = 1$$

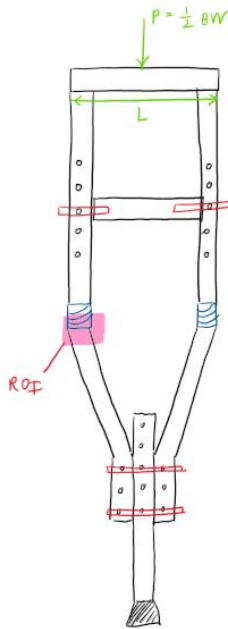
$$K_e = 0.62$$

$$S_e = (0.62)(0.825)(296.5) = 151.678 \text{ MPa}$$

⑥ determine cycles to failure

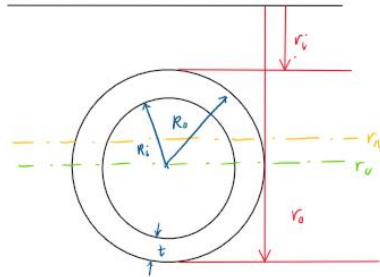
$$n_f = \left(\frac{S_a}{S_e} + \frac{S_m}{S_{ut}} \right)^{-1} = \left(\frac{14.273}{151.678} + \frac{14.273}{593} \right)^{-1} = 8.462$$

4. Curved Beam in Bending



Assumptions:

- a) section is experiencing bending only
- b) cross-section is circular
- c) neutral axis does not shift



$$M = P \cdot L/2 = \frac{1}{4} BW \cdot L = \frac{1}{4} (1000)(0.18) = 45 \text{ N}\cdot\text{m}$$

KNOWN VARIABLES / DIMENSIONS: $BW, L, d_o, t, r_i, S_{ut}$

$$\begin{aligned} r_i &= 40 \text{ mm} & d_o &= 4.5 \text{ cm} & R_o &= 2.25 \text{ cm} \\ d_i &= 4.1 \text{ cm} & R_i &= 2.05 \text{ cm} \end{aligned}$$

① calculate r_c

$$r_c = r_i + R_o = 0.04 \text{ m} + 0.0225 \text{ m} = 0.0625 \text{ m}$$

* r_c is the same for both circles, the centroids overlap.

② calculate r_n

$$\begin{aligned} \int \frac{dA}{r} &= \int \frac{dA}{r} \Big|_{\text{big circle}} - \int \frac{dA}{r} \Big|_{\text{small circle}} \\ &= 2\pi (r_c - \sqrt{r_c^2 - R_o^2}) - 2\pi (r_c - \sqrt{r_c^2 - R_i^2}) \\ &= 2\pi (\sqrt{r_c^2 - R_i^2} - \sqrt{r_c^2 - R_o^2}) \end{aligned}$$

$$\begin{aligned} r_n &= \frac{A}{\int \frac{dA}{r}} = \frac{\pi (R_o^2 - R_i^2)}{2\pi (\sqrt{r_c^2 - R_i^2} - \sqrt{r_c^2 - R_o^2})} = \frac{\pi ((0.0225)^2 - (0.0205)^2)}{2\pi (\sqrt{0.0625^2 - 0.020^2} - \sqrt{0.0625^2 - 0.0225^2})} \\ &= 0.05873 \text{ m} \end{aligned}$$

③ calculate maximum stress

$$\begin{aligned} \tau_{\max} &= \frac{M c_i}{A e r_i} = \frac{(45)(0.01873)}{(2.7 \times 10^{-4})(0.00377)(0.04)} & c_i &= r_n - r_i & e &= r_c - r_n \\ &= 20.7 \text{ MPa} & &= 0.05873 - 0.04 = 0.01873 & &= 0.0625 - 0.05873 = 0.00377 \end{aligned}$$

④ calculate τ_m and τ_a

$$\tau_{\min} = 0 \quad \tau_{\max} = \frac{M c_i}{A e r_i} = 20.7 \text{ MPa}$$

$$\tau_m = \tau_a = \frac{1}{2} \tau_{\max} = 10.35 \text{ MPa}$$

⑤ calculate S_e and S_e

$$S_e' = \frac{1}{2} S_{ut} = \frac{1}{2} (593 \text{ MPa}) = 296.5 \text{ MPa}$$

calculate Marin factors:

$$k_a = 1$$

$$k_b = 1.24 d_o^{-0.107} = 1.24 (45)^{-0.107} = 0.825$$

$$k_b = \begin{cases} (d/0.3)^{-0.107} = 0.879d^{-0.107} & 0.11 \leq d \leq 2 \text{ in} \\ 0.91d^{-0.157} & 2 < d \leq 10 \text{ in} \\ (d/7.62)^{-0.107} = 1.24d^{-0.107} & 2.79 \leq d \leq 51 \text{ mm} \\ 1.51d^{-0.157} & 51 < d \leq 254 \text{ mm} \end{cases} \quad (6-20)$$

$$k_c = 1 \text{ (bending)}$$

$$k_c = \begin{cases} 1 & \text{bending} \\ 0.85 & \text{axial} \\ 0.59 & \text{torsion} \end{cases} \quad (6-26)$$

$$k_d = 1 \text{ (no temperature effects)}$$

$$k_e = 0.620 \text{ (highest reliability)}$$

Table 6-5

Reliability Factors k_e
Corresponding to
8 Percent Standard
Deviation of the
Endurance Limit

Reliability, %	Transformation Variate z_o	Reliability Factor k_e
50	0	1.000
90	1.288	0.897
95	1.645	0.868
99	2.326	0.814
99.9	3.091	0.753
99.99	3.719	0.702
99.999	4.265	0.659
99.9999	4.753	0.620

$$k_f = 1 \text{ (no miscellaneous effects)}$$

$$S_e = k_a k_b k_c k_d k_e k_f S_e' = (0.825)(0.620)(296.5) = 151.657 \text{ MPa}$$

⑥ determine safety factor

$$n_f = \left(\frac{\bar{\sigma}_a}{S_e} + \frac{\bar{\sigma}_m}{S_{ut}} \right)^{-1} = \left(\frac{10.35}{151.657} + \frac{10.35}{593} \right)^{-1} = 11.67$$

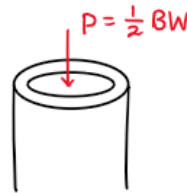
5. Column Buckling



Known Values & Dimensions:

- BW = 1000 N (body weight force on crutch)
- $l = 1.5$ m (length of crutch)
- $d_o = 4.5 \times 10^{-2}$ m (outer diameter of crutch)
- $t = 2 \times 10^{-3}$ m (thickness of crutch wall)
- $d_i = d_o - 2t = 4.1 \times 10^{-2}$ m (inner diameter of crutch)
- $r_o = d_o/2 = 2.25 \times 10^{-2}$ m (outer radius of crutch)
- $r_i = d_i/2 = 2.05 \times 10^{-2}$ m (inner radius of crutch)

consider this area of the crutch for column buckling



Assumptions:

- column is a hollow cylindrical pipe
↳ with same dimensions as crutch pole cross-section
- boundary conditions are fixed/free $\Rightarrow C = 1/4$ *most conservative

Material Properties: (for 7075 Aluminum Alloy T6)

- $E = 71.7$ GPa
- $S_y = 542$ MPa

Check if it is a long or intermediate beam:

↳ find critical slenderness ratio $(\ell/k)_c$:

$$\left(\frac{\ell}{k}\right)_c = \left(\frac{2\pi^2 CE}{S_y}\right)^{1/2} = \left(\frac{2\pi^2 (1/4)(71.7 \times 10^9)}{542 \times 10^6}\right)^{1/2} = 25.55$$

↳ find $\frac{\ell}{k}$ based on the dimensions/geometry:

$$k = \sqrt{\frac{I}{A}}, \quad I = \frac{\pi}{4}(r_o^4 - r_i^4) = \frac{\pi}{4}[(2.25 \times 10^{-2} \text{ m})^4 - (2.05 \times 10^{-2} \text{ m})^4]$$

$$I = 6.258 \times 10^{-8} \text{ m}^4$$

$$A = \pi(r_o^2 - r_i^2) = \pi[(2.25 \times 10^{-2} \text{ m})^2 - (2.05 \times 10^{-2} \text{ m})^2]$$

$$A = 2.702 \times 10^{-4} \text{ m}^2$$

$$\Rightarrow k = \sqrt{\frac{6.258 \times 10^{-8}}{2.702 \times 10^{-4}}} = 0.0152 \text{ m}$$

$$\Rightarrow \frac{\ell}{k} = \frac{1.5 \text{ m}}{0.0152 \text{ m}} = 98.56$$

$\therefore$ we have $\frac{\ell}{k} > \left(\frac{\ell}{k}\right)_c$, so it is a long column & can use the Euler formula

Critical Buckling Load from Euler's formula: $P_{cr} = \frac{C\pi^2 EI}{\ell^2}$

$$P_{cr} = \frac{(1/4)\pi^2 (71.7 \times 10^9)(6.258 \times 10^{-8})}{1.5^2} = 4920.53 \text{ N}$$

$$\text{Safety Factor: } n = \frac{P_{cr}}{P} = \frac{4920.53 \text{ N}}{\frac{1}{2}(1000 \text{ N})} = 9.84$$

Appendix D: Stress Analysis MATLAB Code

`%% Curved Beam in Bending`

`clc; clear all; close all;`

`% Crutch dimensions`

`BW = 1000; %Body weight, N`

`W = 18e-2; %Width of crutch, m`

`do = 4.5e-2; %Outer diameter of crutch, m`

`t = 2e-3; %Thickness of crutch wall, m`

`ri = 40e-3; %Radius of curvature, m`

`% Material properties`

`Sut = 593e6; %Ultimate strength, Pa`

`% Define dependent variables`

`Ro = do/2; %Inner radius of crutch`

`di = do-2*t; %Inner diameter of crutch`

`Ri = di/2; %Inner radius of crutch`

`% Calculate moment`

`M = 0.25*BW*W;`

`% Calculate rc`

`rc = ri + Ro;`

`fprintf('Radius of centroid, rc: %d \n',rc)`

`% Calculate rn`

`int_A = 2*pi*((rc^2-Ri^2)^0.5-(rc^2-Ro^2)^0.5);`

`A = pi*(Ro^2-Ri^2); %Area of cross section`

`rn = A/int_A;`

`fprintf('Radius of neutral axis, rn: %d \n',rn)`

`% Calculate maximum stress`

`ci = rn-ri;`

`e = rc-rn;`

`Smax = M*ci/(A*e*ri);`

`fprintf('Maximum stress, Smax: %d \n',Smax)`

`% Calculate mean stress and stress amplitude`

`Sa = Smax/2;`

`Sm = Smax/2;`

`% Calculate endurance limit`

`Se_p = 0.5*Sut;`

`ka = 1;`

`kb = 1.24*(do*10^3)^-0.107;`

`kc = 1;`

`kd = 1;`

`ke = 0.62;`

`kf = 1;`

`Se = ka*kb*kc*kd*ke*kf*Se_p;`

`fprintf('Endurance limit, Se: %d \n',Se)`

```

% Determine safety factor
nf = (Sa/Se + Sm/Sut)^-1;
fprintf('Safety factor, nf: %d \n',nf)

% Determine cycles to failure
f = 1.06 - 0.00041*(Sut*10^-6) + (1.5e-7)*(Sut*10^-6)^2;
a = ((f*Sut)^2)/Se;
b = (-1/3)*log(f*Sut/Se);
Sar = Sa/(1-(Sm/Sut));
N = (Sar/a)^(1/b);
fprintf('Cycles to failure, N: %d \n',N)

%% Transverse Holes in Bending

clc; clear all; close all;

% Crutch dimensions
BW = 1000; %Body weight, N
W = 18e-2; %Width of crutch, m
do = 4.5e-2; %Outer diameter of crutch, m
t = 2e-3; %Thickness of crutch wall, m
a = 2.5e-3; %Radius of transverse hole

% Material properties
Sut = 593e6; %Ultimate strength, Pa

% Define dependent variables
di = do-2*t; %Inner diameter of crutch

% Calculate moment
M = 0.25*BW*W;

% Calculate stress concentration factor
c1 = 3;
c2 = -6.25-0.585*(di/do)+3.115*(di/do)^2;
c3 = 41 - 1.071*(di/do)-6.746*(di/do)^2;
c4 = -45 + 1.389*(di/do) + 13.889*(di/do)^2;
Kt = c1 + c2*2*a/do + c3*(2*a/do)^2 + c4*(2*a/do)^3;
%Kt = 2.4
fprintf('Stress concentration factor, Kt: %d\n', Kt)

% Calculate nominal stress
S0 = 32*M*do/(pi*(do^4-di^4));
fprintf('Nominal stress, S0: %d \n',S0)

% Calculate mean stress and stress amplitude
Sa = Kt*S0/2;
Sm = Kt*S0/2;

% Calculate endurance limit
Se_p = 0.5*Sut;
ka = 1;
kb = 1.24*(do*10^3)^-0.107;
kc = 1;

```

```

kd = 1;
ke = 0.62;
kf = 1;
Se = ka*kb*kc*kd*ke*kf*Se_p;
fprintf('Endurance limit, Se: %d \n',Se)

% Determine safety factor
nf = (Sa/Se + Sm/Sut)^-1;
fprintf('Safety factor, nf: %d \n',nf)

% Determine cycles to failure
f = 1.06 - 0.00041*(Sut*10^-6) + (1.5e-7)*(Sut*10^-6)^2;
a = ((f*Sut)^2)/Se;
b = (-1/3)*log(f*Sut/Se);
Sar = Sa/(1-(Sm/Sut));
N = (Sar/a)^(1/b);
fprintf('Cycles to failure, N: %d \n',N)

%% Threaded Beam in Axial Loading

clc; clear all; close all;

% Crutch dimensions
BW = 1000; %Body weight, N
p = 2e-3; %Pitch, mm
nt = 4; %Number of turns
do = 4.5e-2; %Outer diameter of crutch, m

% Material properties
Sut = 593e6; %Ultimate strength, Pa

% Define dependent variables
dr = 1.226869*p; %Diameter of thread
V = 0.25*BW; %Axial load, N

% Calculate axial stress
Saxial = -4*V/(pi*dr^2);
fprintf('Axial stress, Saxial: %d \n',Saxial)

% Calculate bending stress
Sbending = 6*V/(pi*dr*nt*p);
fprintf('Bending stress, Sbending: %d \n',Sbending)

% Calculate total stress, mean stress, and stress amplitude
Smax = abs(Saxial+Sbending);
Sa = Smax/2;
Sm = Smax/2;

% Calculate endurance limit
Se_p = 0.5*Sut;
ka = 1;
kb = 1.24*(do*10^3)^-0.107;
kc = 1;
kd = 1;
ke = 0.62;

```

```

kf = 1;
Se = ka*kb*kc*kd*ke*kf*Se_p;
fprintf('Endurance limit, Se: %d \n',Se)

% Determine safety factor
nf = (Sa/Se + Sm/Sut)^-1;
fprintf('Safety factor, nf: %d \n',nf)

% Determine cycles to failure
f = 1.06 - 0.00041*(Sut*10^-6) + (1.5e-7)*(Sut*10^-6)^2;
a = ((f*Sut)^2)/Se;
b = (-1/3)*log(f*Sut/Se);
Sar = Sa/(1-(Sm/Sut));
N = (Sar/a)^(1/b);
fprintf('Cycles to failure, N: %d \n',N)

%% Pin shear

clc; clear all; close all;

% Crutch dimensions
BW = 1000; %Body weight, N
D = 5e-3; %Diameter of pin/bolt

% Material properties
Sut = (324e6)/2; %Ultimate strength, Pa

% Calculate transverse stress
Smax = BW/(4*(pi*0.25*D^2));
fprintf('Transverse stress, Smax: %d \n',Smax)

% Calculate total stress, mean stress, and stress amplitude
Sa = Smax/2;
Sm = Smax/2;

% Calculate endurance limit
Se_p = 0.5*Sut;
ka = 1;
kb = 1.24*(D*10^3)^-0.107;
kc = 1;
kd = 1;
ke = 0.62;
kf = 1;
Se = ka*kb*kc*kd*ke*kf*Se_p;
fprintf('Endurance limit, Se: %d \n',Se)

% Determine safety factor
nf = (Sa/Se + Sm/Sut)^-1;
fprintf('Safety factor, nf: %d \n',nf)

% Determine cycles to failure
f = 1.06 - 0.00041*(Sut*10^-6) + (1.5e-7)*(Sut*10^-6)^2;
a = ((f*Sut)^2)/Se;
b = (-1/3)*log(f*Sut/Se);
Sar = Sa/(1-(Sm/Sut));

```

```

N = (Sar/a)^(1/b);
fprintf('Cycles to failure, N: %d \n',N)

%% Check for Column Buckling

clc; clear all; close all;

BW = 0.5*1000; %Body weight, N

do = 4.5e-2; %Outer diameter of crutch, m
t = 2e-3; %Thickness of crutch wall, m

Ro = do/2; %Inner radius of crutch
di = do-2*t; %Inner diameter of crutch
Ri = di/2; %Inner radius of crutch

l = 1.5; % length of crutch, m
C = 1/4; % constant for 1 end fixed (pinned) and 1 end free
E = 71.7e9;
I = (pi/4)*(Ro^4-Ri^4);
A = pi*(Ro^2-Ri^2);
Sy = 542e6;

lk_crit = sqrt((2*C*(pi^2)*E)/Sy);
fprintf('Critical Slenderness Ratio, l/k: %d \n',lk_crit)

lk = 1/(sqrt(I/A));
fprintf('Slenderness Ratio, l/k: %d \n',lk)

P_cr = (C*(pi^2)*E*I)/(l^2); % use Euler eqn bc long column
fprintf('Critical Buckling Load, P_cr: %d \n',P_cr)

nf = P_cr/BW;
fprintf('Safety factor, nf: %d \n',nf)

```

Appendix